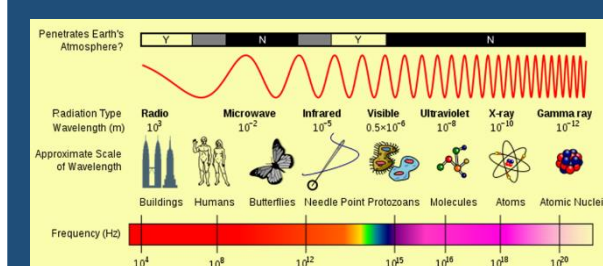
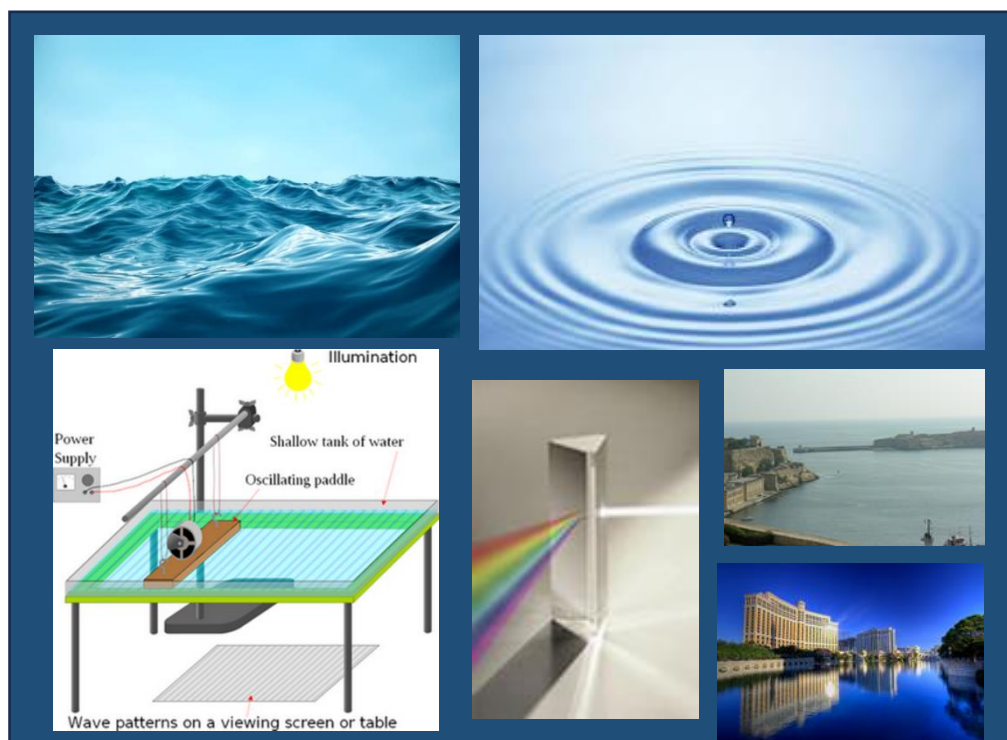


Year 10 Notes - Booklet 1

Waves

Ms Karen



Index

All kinds of waves	Pg 2
Sound Waves	Pg 18
Electromagnetic Waves	Pg 30
Light Waves (Optics)	Pg 34
Worksheets	Pg 69

Waves and energy

What connects waves to energy? **Waves transport energy from place to place.**

A water wave transports Kinetic energy and sound for example.

We can see water waves on the surface. We actually see the **wave crest** (wave top, ripple in water) . The wave crest is also a **wavefront**



Each wavefront represents 1 wave.

Some waves are similar while others are not – there are different types of waves on earth and in space.

The Wave medium

A medium is a material (**matter**) through which a wave travels.

What are waves made of?

Some waves, like water waves, are made of the medium itself.

Example – water is a medium for water waves

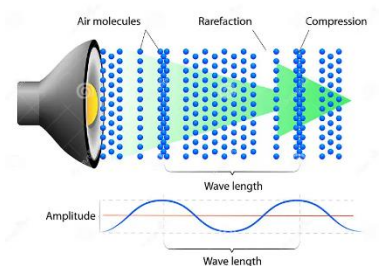


Another example...a slinky wave is made out of the steel springs in the slinky.



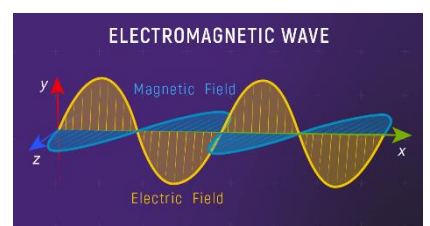
Another example, sound waves travelling in air are made out of air particles

SOUND WAVES



Other waves, such as light waves and radio waves are just made of electric and magnetic force fields. They are made out of pure energy and do not need a medium to travel through.

ELECTROMAGNETIC WAVE



How Do Waves Form?

A vibration /disturbance causes a wave to form.

What is the difference between a vibration and a wave?

A vibration is a wiggle in time

A wave is a wiggle in time and space.



Mechanical Waves

A **mechanical** wave needs a medium through which to travel – it cannot travel through a vacuum.

Examples of mechanical waves

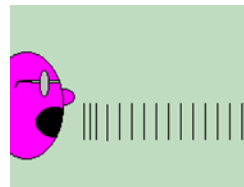
- A slinky wave needs its coils to travel



- Ocean waves need water

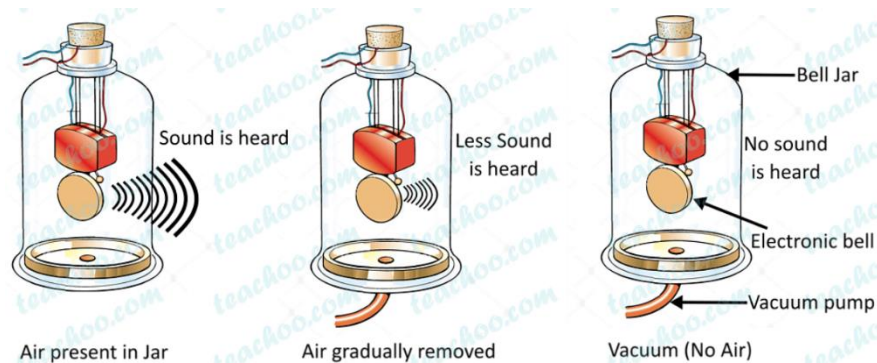


- Sound needs air, or any other medium



The Bell-Jar experiment demonstrates that mechanical waves need a medium to travel through.

<http://www.youtube.com/watch?v=ce7AMJdq0Gw>

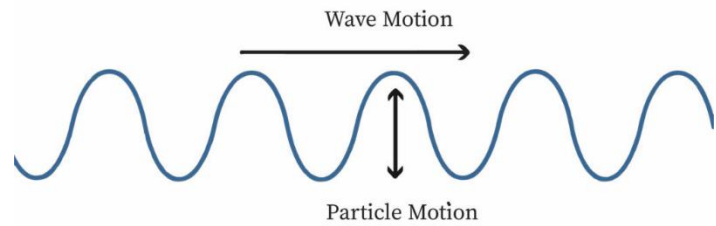


Precautions taken : _____

Particle Motion in Waves

Look at the interactive animations on the powerpoint slide (slide no 10) and keep an eye on 1 particle only.

Every particle is vibrating about a fixed position
It always goes back to where it started. This means that.....



Waves transfer energy without displacing the matter

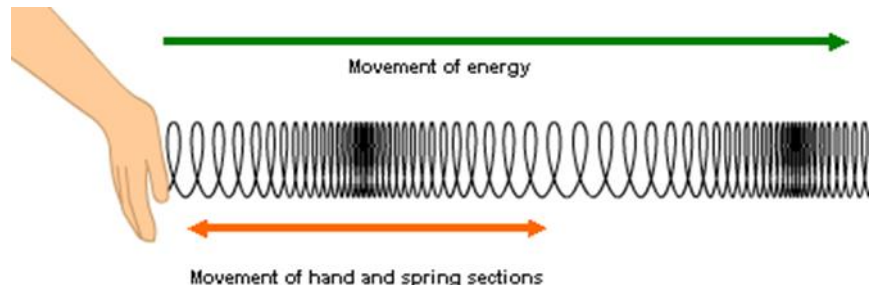
The term displaced means that an object ends up some distance away from its starting position. Since the particles are always going back to their starting position there is no displacement.

Two Types of Mechanical Waves Transverse Waves and Longitudinal Waves

(There are other types but these are not in our syllabus)

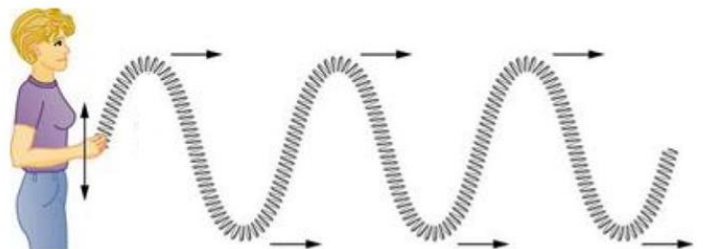
Longitudinal wave (such as sound)

Medium and energy move in parallel.
Parts of the medium are compressed while other parts are stretched, in a pattern.



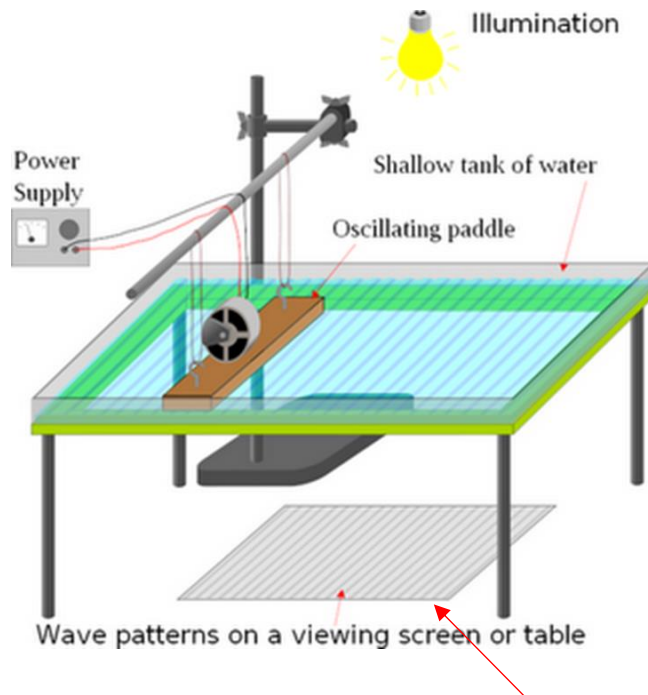
Transverse wave (such as a water wave)

Medium and energy move perpendicular (at 90°) to each other.
Parts of the medium form crests (peaks) while other parts form troughs (depressions).



Properties of Waves

Scientists study wave properties by Producing water waves in a **Ripple Tank**



Water waves have crests and troughs it is easy for us to view the shadow of the wavefronts to see how the waves behave.

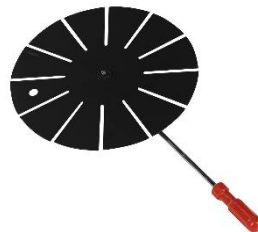
When we want to draw a diagram to show how a wave behaves when entering another medium, for example, we draw a number of wavefronts to represent the wave, like this.

The Importance of a Stroboscope to view the wave pattern in the ripple tank

A stroboscope is a disc containing slits which allow us to see through them while the stroboscope spins.

When we view the continuously moving wave pattern through the stroboscope **the pattern seems to freeze** because we only see it when there is a slit.

This makes it easier for us to observe and trace the wave pattern.



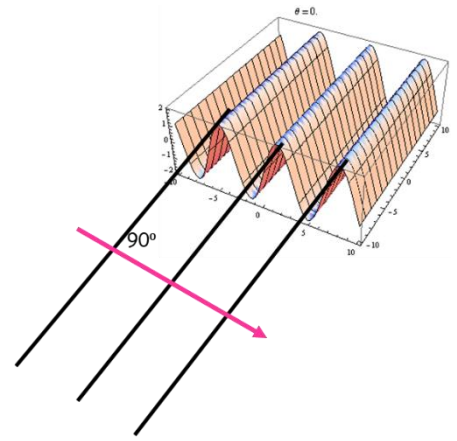
Hand stroboscope



digital stroboscope

How to draw ray diagrams to represent a wave

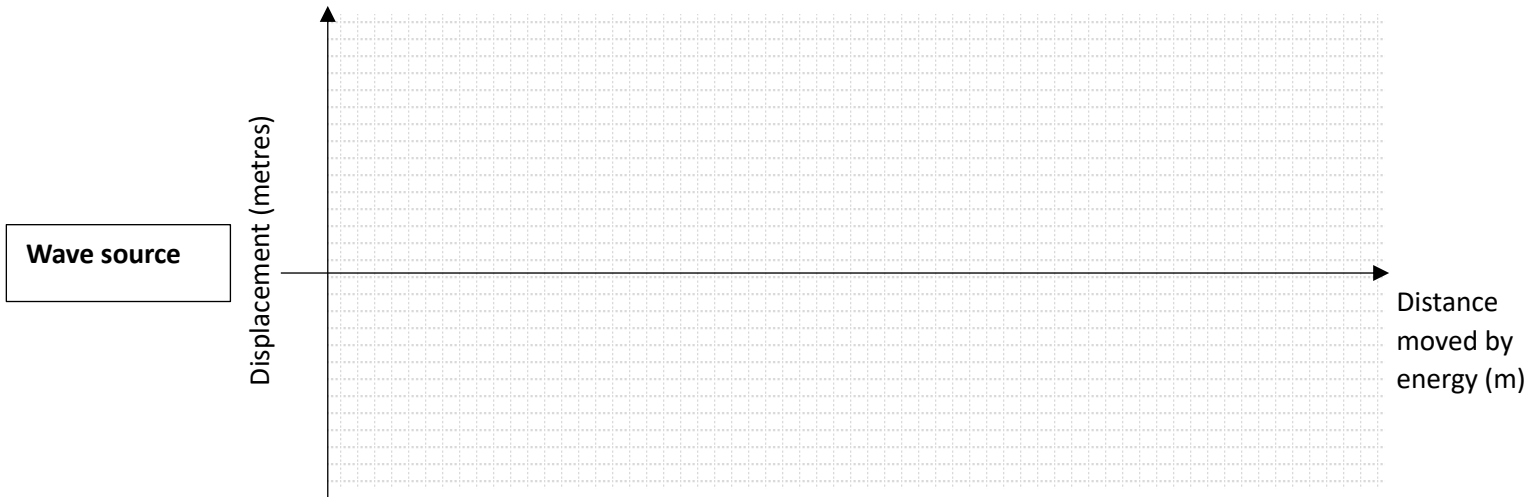
1. Observe the direction in which the waves are moving.
2. Draw the wavefronts – 3 are usually enough. These are 3 parallel lines equidistant from each other. The distance between each wavefront is the wavelength of each wave.
3. Draw the ray that represents the wave and its direction.
4. The ray is drawn perpendicular to the wavefronts



Wave anatomy

Draw a transverse wave and a longitudinal wave below, and mark the following on these waves: **Wave direction**, **crest**, **trough**, **compression**, **rarefaction**, **amplitude** (*only possible on transverse wave*), **wavelength**

Transverse wave



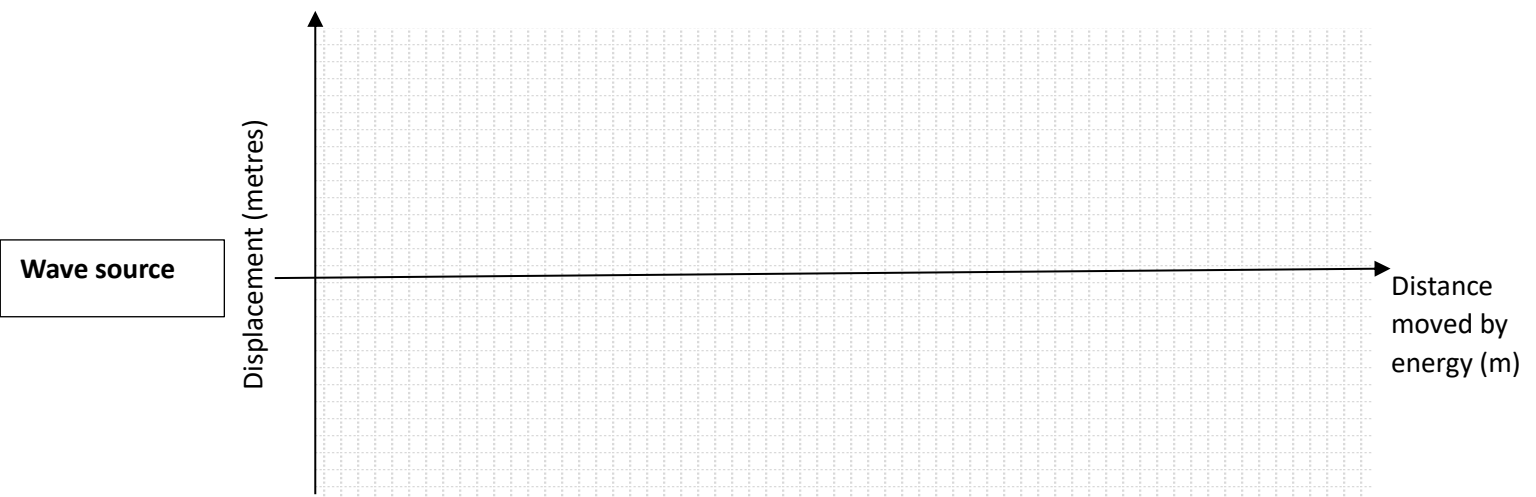
Crest : _____

Trough : _____

Wavelength: _____ symbol : __

Amplitude : _____ symbol: __

Longitudinal wave



Compression : _____

Rarefaction : _____

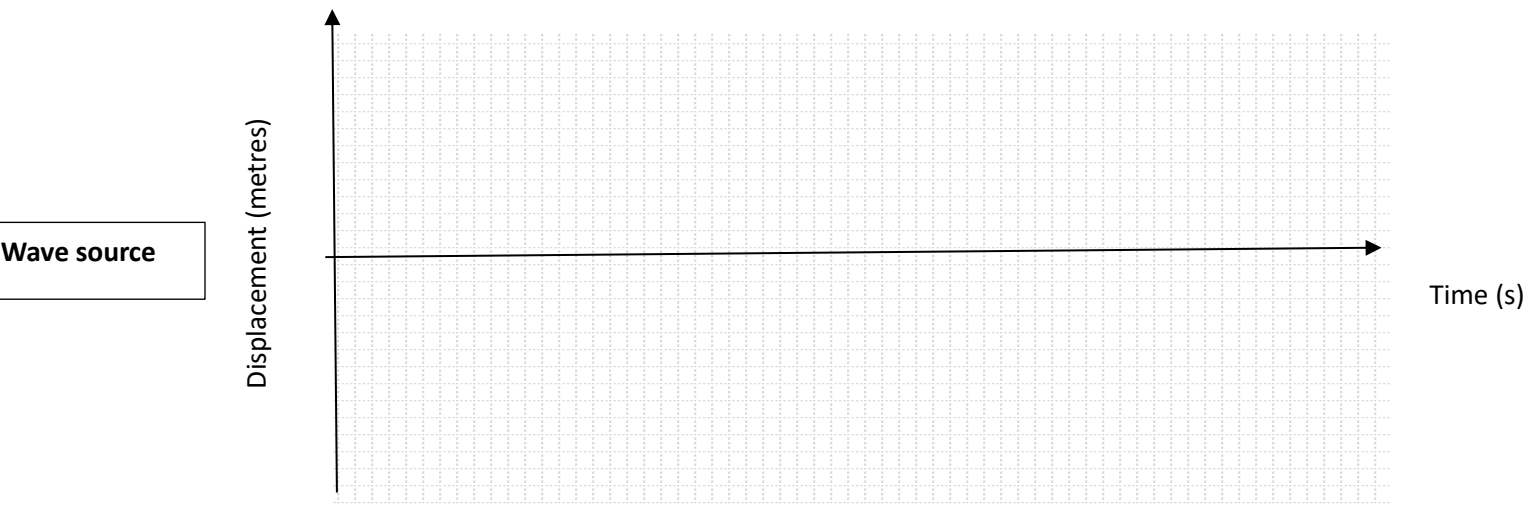
Wavelength: _____ symbol : __

Amplitude : _____ symbol: __

Draw a transverse wave and a longitudinal wave below, and mark the following on these waves:

Wave direction, amplitude(only possible on transverse wave) ,**frequency, periodic time**

Transverse wave



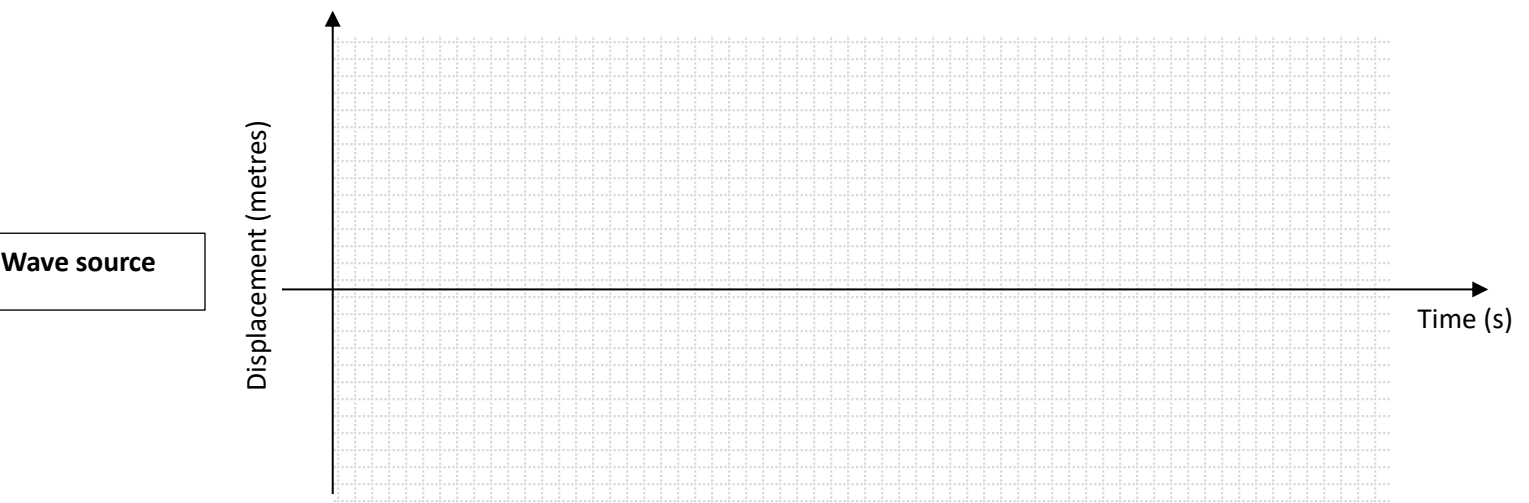
Amplitude : _____ symbol : ____

Frequency: _____ symbol : ____

How to find frequency from the graph: _____

Periodic time : _____ symbol: ____

Longitudinal wave



Amplitude : _____ symbol : ____

Frequency: _____ symbol : ____

Periodic time : _____ symbol: ____

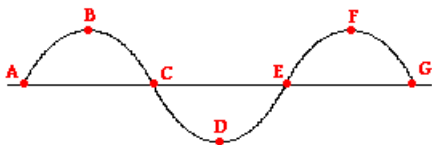
Wave properties worksheet 1

Wave Terms to memorise

Mechanical, wave medium, transverse, longitudinal, wavefront, crest, trough, compression, rarefaction, displacement, amplitude, wavelength (λ), frequency,

- Which one of the statements below is **not** a characteristic of all mechanical waves? Choose one answer.
 - ☐ a. They transport energy
 - ☒ b. They travel in a direction that is at right angles to the direction of the particles in the medium
 - ☐ c. They are created by a vibrating source
 - ☐ d. They consist of disturbances or oscillations of a medium
- In order for Jack to hear Jill, air molecules must move from the lips of Jill to the ears of John.
 - ☐ True ☐ False
- The sonar device on a fishing boat uses underwater sound to locate fish. Would you expect sonar to be a longitudinal or transverse wave?
 - ☐ a. Transverse wave
 - ☒ b. Longitudinal wave
- A science fiction film depicts inhabitants of one spaceship (in outer space) hearing the sound of a nearby spaceship as it zooms past at high speeds. Is this fact or fiction?
 - ☐ a. Fiction
 - ☒ b. Fact

5. The diagram below represents a transverse wave



Indicate the interval that represents the wavelength

Circle one answer

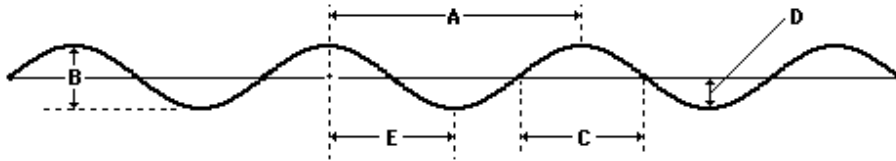
- C-G • B-D

6. A wave is transverse if the particles of the medium are moving back and forth in a leftward and rightward direction. This type of wave is known as a ____.

Choose one answer.

- ☐ a. Transverse wave
- ☒ b. Mechanical wave
- ☐ c. Longitudinal wave
- ☒ d. Electromagnetic waves

7.



- a. Which letter indicates the amplitude? _____
- b. Which letter measures the wavelength of this wave? _____

8. Two waves are traveling through a container of an inert gas. Wave A has an amplitude of .1 cm. Wave B has an amplitude of 0.2 cm. The energy transported by wave B must be _____ the energy transported by wave A. Choose one answer.

- ☐ the same as
- ☒ greater than
- ☐ less than

9. As the frequency of the wave increases, its periodic time _____ Choose one answer.

- ☐ remains the same
- ☒ decreases
- ☐ increases

10. Sound is a mechanical wave. This means that : Choose one answer.

- ☐ a. Particles of the medium move perpendicular to the direction of energy transport
- ☒ b. Particles of the medium regularly and repeatedly oscillate about their rest position
- ☐ c. A medium is required in order for the sound wave to transport energy
- ☒ d. A sound wave transports its energy through a vacuum

Frequency and Periodic time formulae

Periodic time is given by : $T = \frac{1}{f}$, or time to complete one wave, unit s (seconds)

Wave frequency is given by : $f = \frac{1}{T}$, or number of waves per second, unit Hz (Hertz)

11. A tennis coach paces back and forth along the sideline 10 times in 2 minutes. Calculate the frequency of her pacing.
- .
-

12. A wave is introduced into a thin wire held tightly at each end. It has an amplitude of 3.8cm, a frequency of 51.2Hz and a distance from a crest to a neighbouring trough of 12.8cm. Determine the periodic time of such a wave.
-

13. Freida the fly flaps its wings back and forth 121 times every second. What is the periodic time of the wing flapping?
-

14. A periodic time of 5s corresponds to a frequency of
-

Wave velocity/speed

Definition :

The velocity/speed with which the wave (energy) travels

Symbol : v

Unit : metres per second, m/s

(Some text books show wave speed with symbol c .

c is actually the symbol for speed of light, light also being a wave)

There are 2 formulae to calculate wave speed

Formula 1:

$$\text{Wave Speed/velocity} = \frac{\text{distance covered by wave}}{\text{time}}$$

Worked example 1

If the crest of an ocean wave moves a distance of 20 m in 10s, what is the speed of the wave?

OR

Formula 2 :

$$v = f\lambda$$

- f = wave frequency , in Hertz
- λ = wavelength, in metres
- v = wave speed , in m/s

Worked example 2

A wave has wavelength 2m and frequency 2 Hz. At what speed is it travelling?

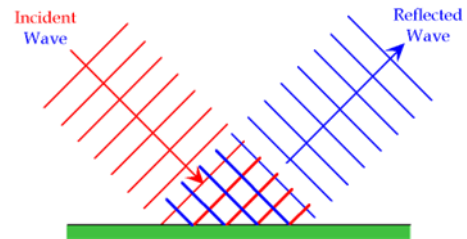
Wave Effects

When a wave reaches a medium boundary, the wave is either....

(The straight lines are wavefronts)

Reflected

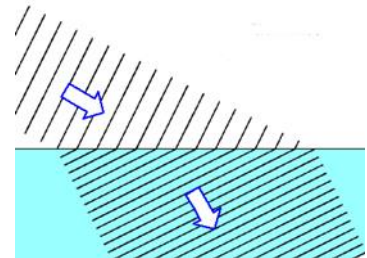
- bounces off barrier
- changes direction
- Keeps same medium, speed, wavelength, frequency
- *Reflection is simulated in a ripple tank by placing smooth steel barriers on its surface for the waves to reflect against. The wavefronts are then observed at the bottom of the tank.*



OR

Refracted

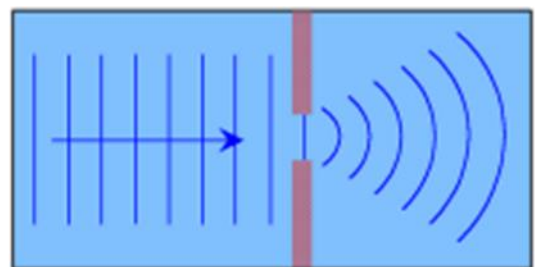
- goes through barrier
- changes medium, speed, wavelength
- May also change direction due to change in speed
- Same frequency
- *Refraction is simulated in a ripple tank by placing a rectangular plastic sheet at the bottom of the surface and varying its angle. The waves will pass over the sheet and refract as they enter/exit the shallow water patch. The wavefronts are then observed at the bottom of the tank.*



OR

Diffraction

- goes through gap or around a corner and spreads out keeps same medium
- Keeps same speed, same wavelength, frequency
- *Diffraction is simulated in a ripple tank by placing steel barriers along a straight line, with a gap inbetween. The width of the gap is varied and the wave behaviour on exit of the gap is observed. The wavefronts are then observed at the bottom of the tank.*

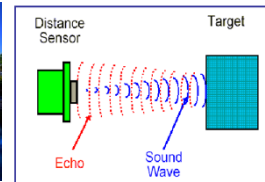


OR

A mixture of any of the above

Wave Reflection – *applies to all wave types*

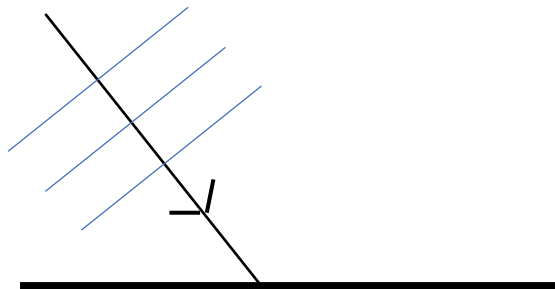
Definition : A change in direction of the waves when they bounce off a medium boundary.



The reflected wave is an exact copy of the incident wave
– same wavelength, same frequency , same speed

Only the direction changes

Simulation of Wave Reflection in a Ripple tank



Complete the ray diagram above by doing the following:

1. Draw the Normal, an imaginary (drawn dotted) line that is drawn at the intersection between ray and reflecting surface (or boundary). This line is drawn perpendicular to the surface.
2. Label the incident wavefronts on the diagram. These are the wavefronts going towards the reflecting surface.
3. Label the incident ray on the diagram.
4. Measure the angle of incidence using a protractor. This is the angle between the incident ray and the Normal. _____. Also label it on the diagram, as angle *i*.
5. Reflect the incident ray about the Normal (just like Mathematical Transformations). This means that you need to construct a new ray at the same angle to the Normal, but on the other side of it.
6. Mark the direction of the reflected ray by drawing an arrow on it.
7. Label the reflected ray.
8. Draw 3 reflected wavefronts.
9. Label the angle of reflection as angle *r*.

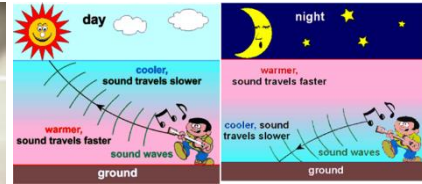
The First Law of Reflection

Angle of incidence = Angle of reflection

The 2nd law will be discussed when we cover Optics

Wave Refraction

– applies to all wave types



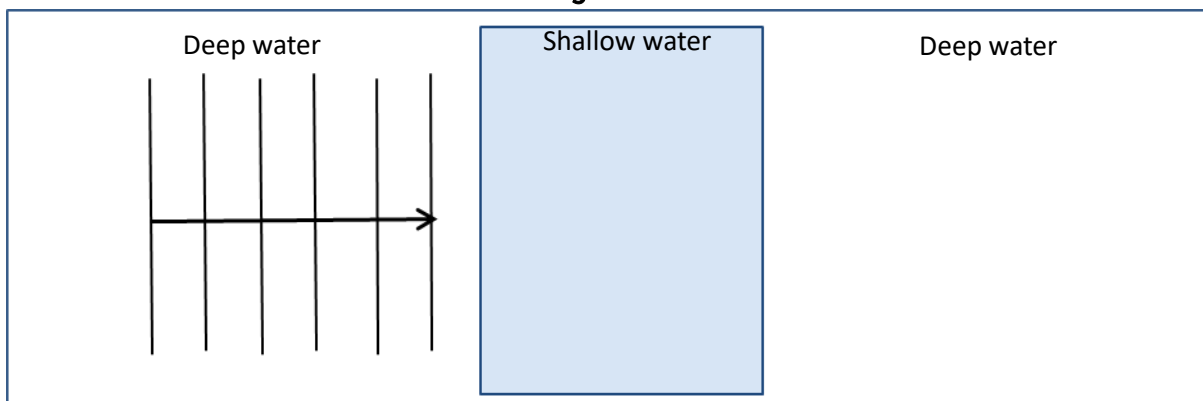
Definition : The wave undergoes a change in speed when it passes from one medium to another

Simulation of Wave Refraction in a Ripple tank

The adjacent diagram shows waves in a ripple tank. By inserting a plastic sheet in the centre we can simulate a change in medium. Note the medium is still water in this case, but this is a good simulation. We can only do this simulation with water waves.

Observe and draw the wavefronts and wave direction as the waves enter the shallow water and as they exit.

Waves enter and exit shallow water at an angle of incidence of 0°



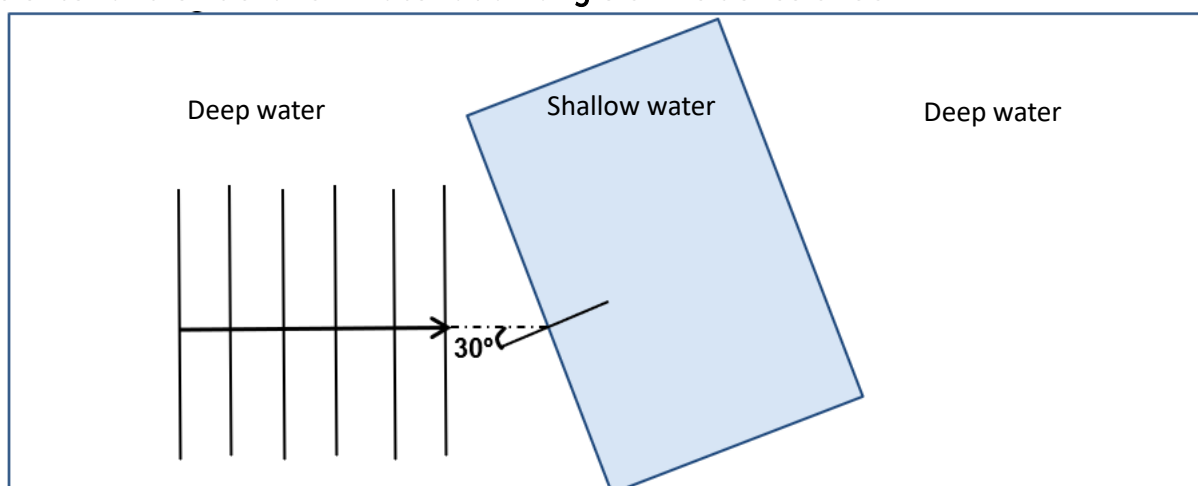
What do we observe to change on entry/exit from the shallow water ?

Change observed	Deep to shallow water	Shallow to deep water
Speed		
Wavelength		
Direction		
Angle of refraction (r)		

Frequency does not change at all – This is determined by the wave source and remains constant.

Observe and draw the wavefronts and wave direction as the waves enter the shallow water and as they exit.

Waves enter and exit shallow water at an angle of incidence of 30°



What do we observe to change on entry/exit from the shallow water :

Change observed	Deep to shallow water	Shallow to deep water
Speed		
Wavelength		
Direction		
Angle of refraction (r)		

Frequency does not change at all – This is determined by the wave source and remains

Note about frequency : Since $v = f\lambda$ the frequency of the wave will only change if the wave speed is constant and the wavelength (λ) changes. This does not apply to any wave refraction.

Conclusion to our Refraction simulation

Wave speed :

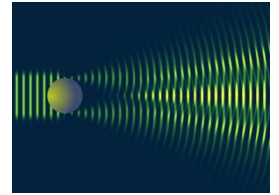
Wavelength :

Wave direction :

Frequency :

Wave Diffraction

– applies to all wave types



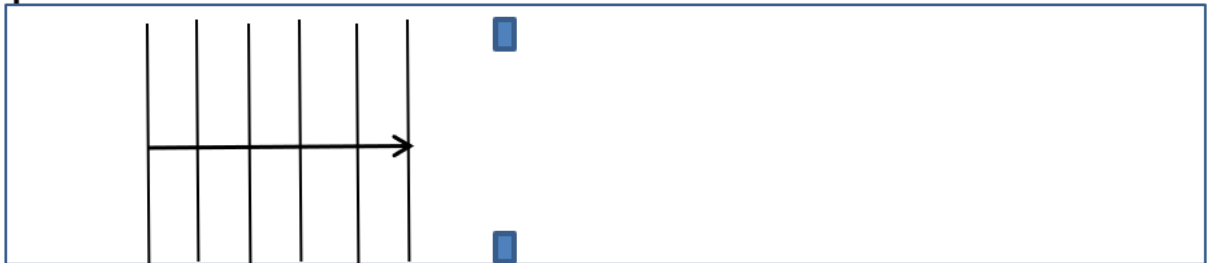
Definition : The spreading out of waves as they pass through a gap or around an obstacle

Simulation of Wave Diffraction in a Ripple tank

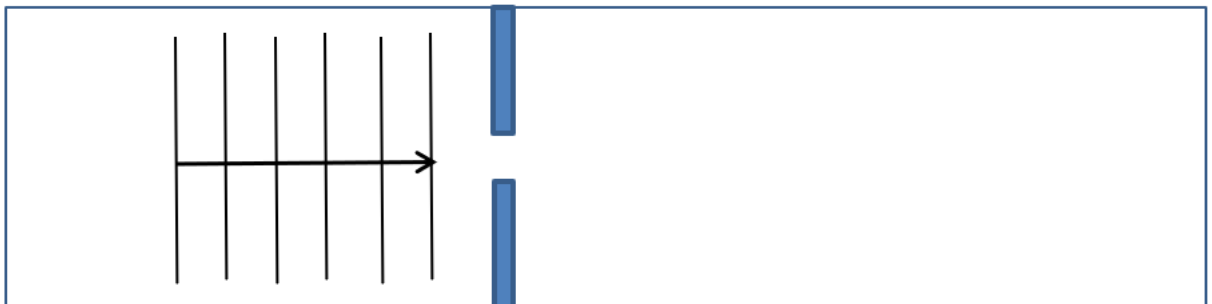
The adjacent diagram shows waves in a ripple tank. By placing barriers in a line with a gap in between we can observe what happens to the water waves when we widen or narrow the gap.

Observe and draw the wavefronts and wave direction as the waves exit the gap or corner.

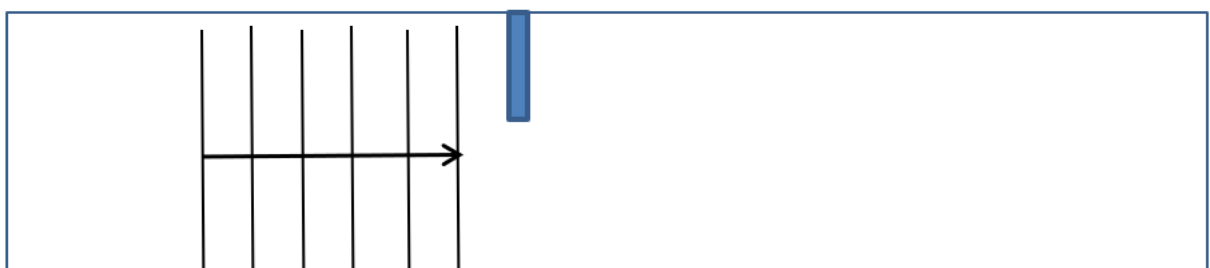
Wide gap



narrow gap



Corner – infinite gap



Conclusion to our Diffraction Simulation

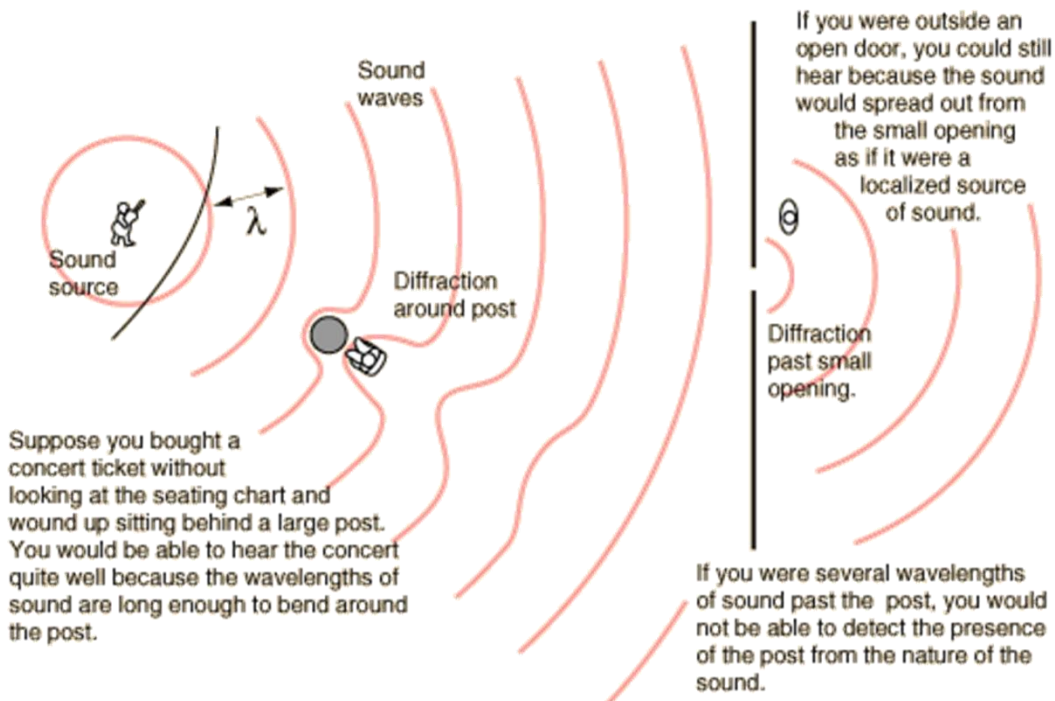
Remember to find the relationship between gap width and wave direction

Wave speed:

Wavelength
and
frequency:

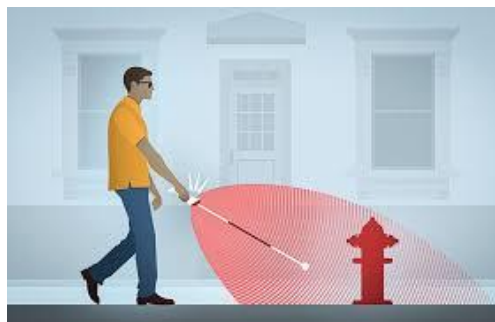
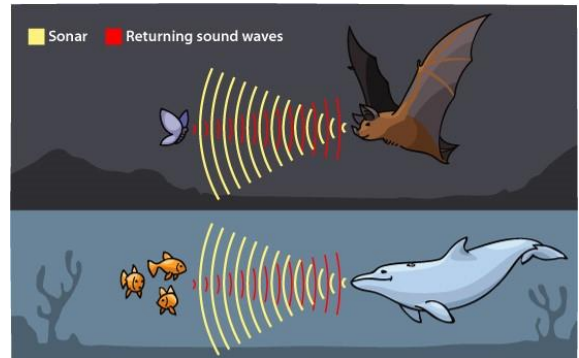
Wave direction:

An example showing how sound waves are diffracted around obstacles at a concert

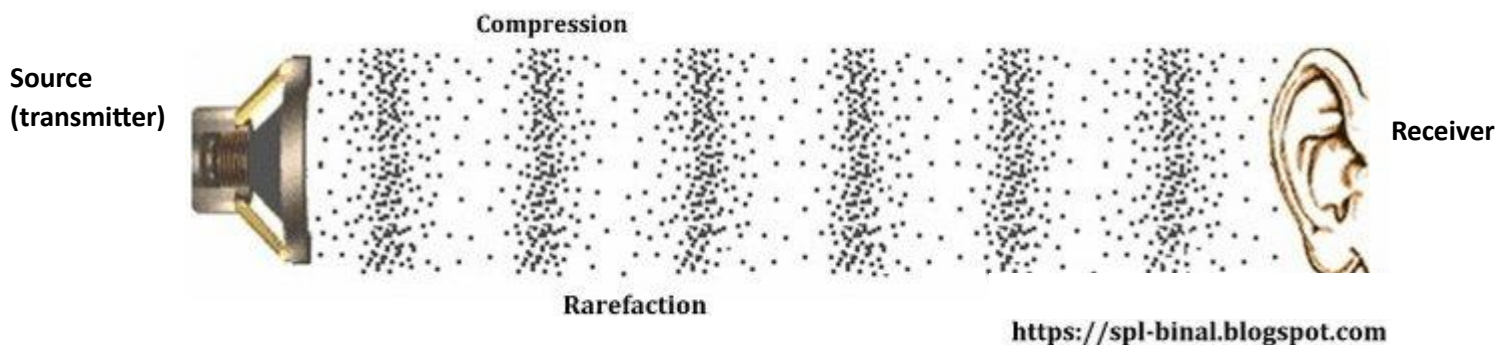


Sound

A mechanical, longitudinal wave that living organisms use to communicate with each other



How is sound produced?



- The vibrating source causes the medium to vibrate too.
- When the source pushes out, the medium particles are compressed (compression).
- When the source pulls in, the particles are pulled back (rarefaction) , thus forming a longitudinal wave.
- Kinetic energy is transmitted to our ears and detected by brain nerves, resulting in our definition of sound.
- The particles in the medium vibrate at the source frequency.

Sound is a wave because...

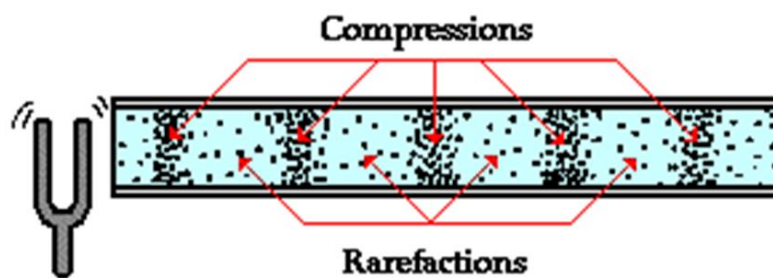
Sound has all the properties of waves

- It transports Kinetic energy from one place to another (the source vibrations are also felt at the receiver end).
- 2. The energy wave originates from a vibrating object
- 3. It has wavelength, amplitude, frequency and periodic time
- The equation $v = f\lambda$ applies to sound waves too

Sound waves – what type of wave are they?

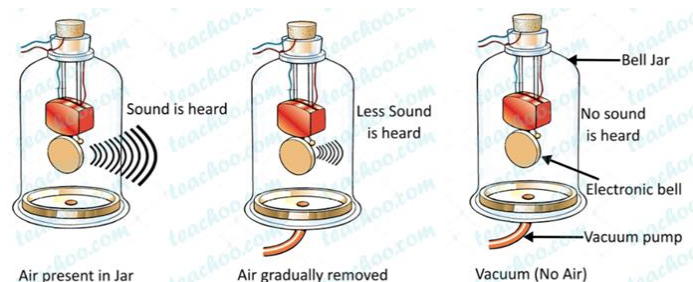
Sound waves are longitudinal waves

Experiments have shown that when a sound is made the particles in the medium oscillate in parallel to the wave direction, therefore we know that the wave motion is longitudinal.



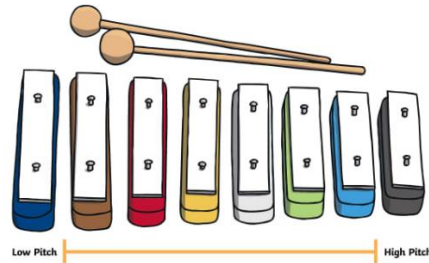
Sound waves are also mechanical waves

We know that sound cannot travel in a vacuum from previous lessons, so sound is a mechanical wave. (Bell Jar Experiment)



Some Properties of Sound

Pitch – our sensation of how high or low a note is



For example, a scream has a high pitch while a fog horn has a low pitch.
The higher the pitch, the higher the frequency

Pitch depends on frequency

It is not frequency

Loudness (sound intensity)

The higher the wave amplitude, the louder the sound.

Since amplitude represents the amount of energy carried by the wave,
loudness also depends on how much energy the wave carries.

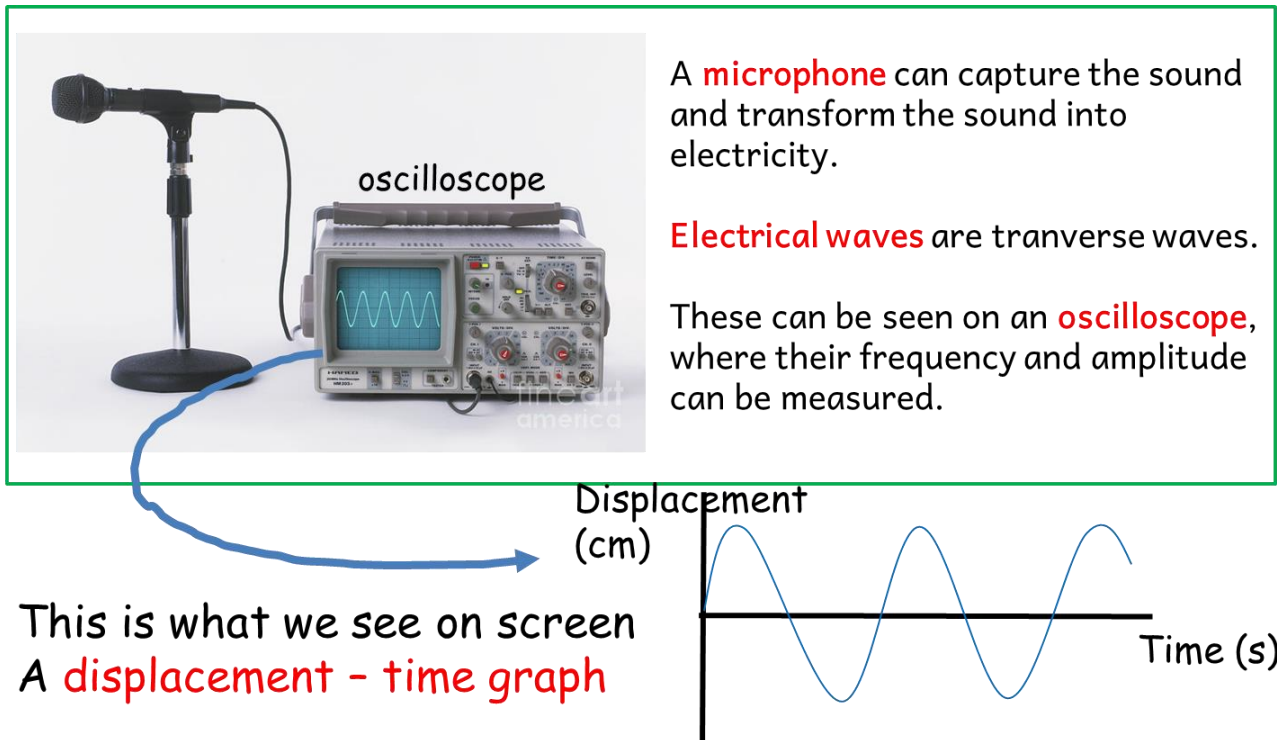
Loudness depends on amplitude

It is not amplitude

Loudness is measured in decibels (dB)

How do we measure the pitch and loudness of a sound?

By measuring the frequency and amplitude of the sound wave



Can sound travel through any medium?

NO!

- **Sound can travel through gases and liquids**
- Sound can also travel through solids, as long as the particles in the medium have elastic properties. They must be able to go back to their original positions after being displaced.
So sound does not travel through plasticine for example, because plasticine is not elastic.

The Speed of Sound

What does the speed of sound depend on?

Like any other wave, its speed depends on the medium in which it is travelling.

Factors affecting speed of sound :

- Sound travels fastest in solids and slowest in gases (molecular bonds are greatest in solids).
- In solids - sound is faster in solids that are less dense, for example sound travels faster in Aluminium rather than iron (denser molecules have more mass and are slower to start moving).
- In liquids and gases – sound travels faster at higher temperatures and humidity (hotter molecules are faster).

No need to study this table

<u>Material</u>	<u>Speed of Sound</u>
Rubber	60 m/s
Air at 20°C	343 m/s
Air at 40°C	355 m/s
Lead	1210 m/s
Sea water 20°C	1522m/s
Gold	3240 m/s
Glass	4540 m/s
Copper	4600 m/s
Aluminum	6320 m/s

Plan an experiment to measure the speed of sound in air.

Required: a drawing of the experiment setup, a list of apparatus, a list of instructions and 2 precautions for accuracy.

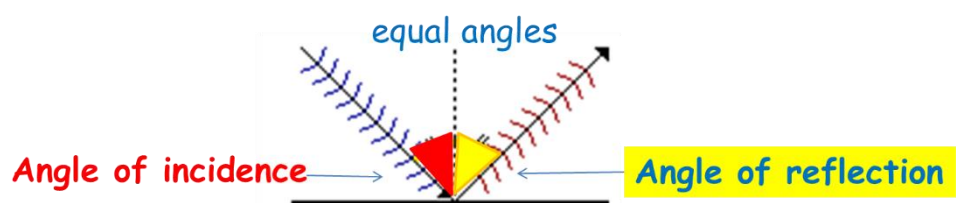
Plan as demonstrated by teacher

Required: a drawing of the experiment setup, a list of apparatus, a list of instructions and 2 precautions for accuracy.

Sound Reflection

We call the reflected wave an echo.

The law of reflection holds also for sound waves



- Good reflectors of sound are rigid and smooth.
- Poor reflectors of sound are soft and rough.
- Some materials absorb all the sound wave and reflect nothing. These are porous materials. These materials are used to insulate the walls of a room, so that no sound escapes to the outside.

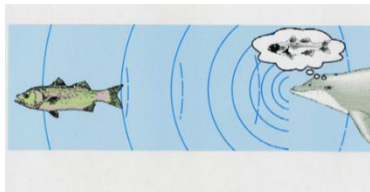
Worked example reflection:

A ship is 220m from a large cliff when it sounds its foghorn. Take the speed of sound in air is 345m/s .

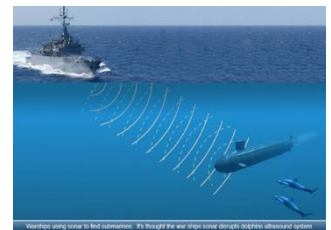
1. How far does the sound travel before the ship can hear the echo of the horn?
2. What time delay is there before the echo is heard?

Why Are Echoes Useful?

Some animals, such as bats (who are blind) and whales and dolphins (who have to navigate in the dark) use echolocation to find their way around.



Ships and boats use echolocation to detect fish or other materials underwater.



Ultrasound medical screening uses **ultrasound** echoes to build images of internal organs or unborn babies.



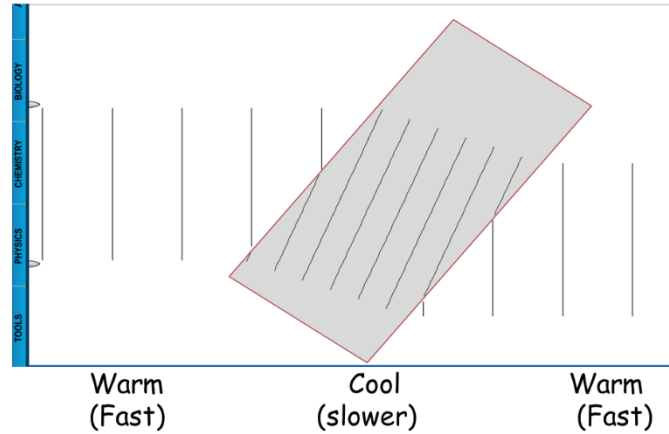
NDT (Non destructive testing) consists of using **ultrasound** echos to locate faults in expensive machinery without breaking it apart



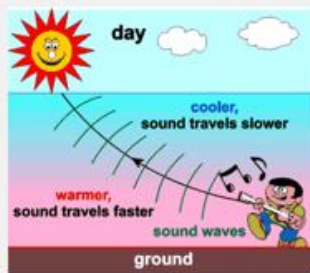
Sound Refraction

When does Sound Refraction occur?

When sound passes from one medium to another – or the same medium but with different properties, for example warm air and cool air.



Examples of Sound Refraction



Refraction due to changes in air temperature - we can't always hear a sound from an object that we can see some distance away from us (sound waves bend)

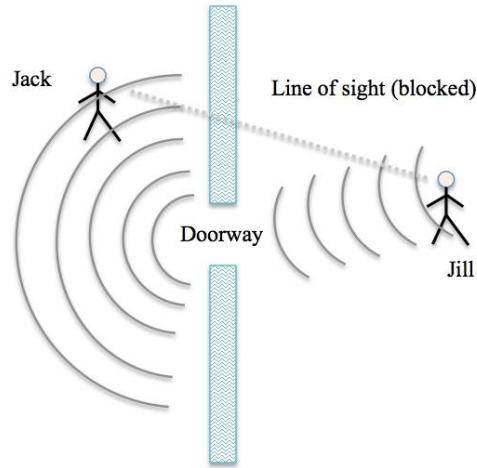


Refraction due to changes in water temperature - A submarines hides in pockets of water having a different temperature so that sonar waves do not reach it due to bending

Sound Diffraction

Why is it, that if a person hides from us around the corner of a building, this person can still hear the sounds we make, even if there are no reflecting surfaces nearby?

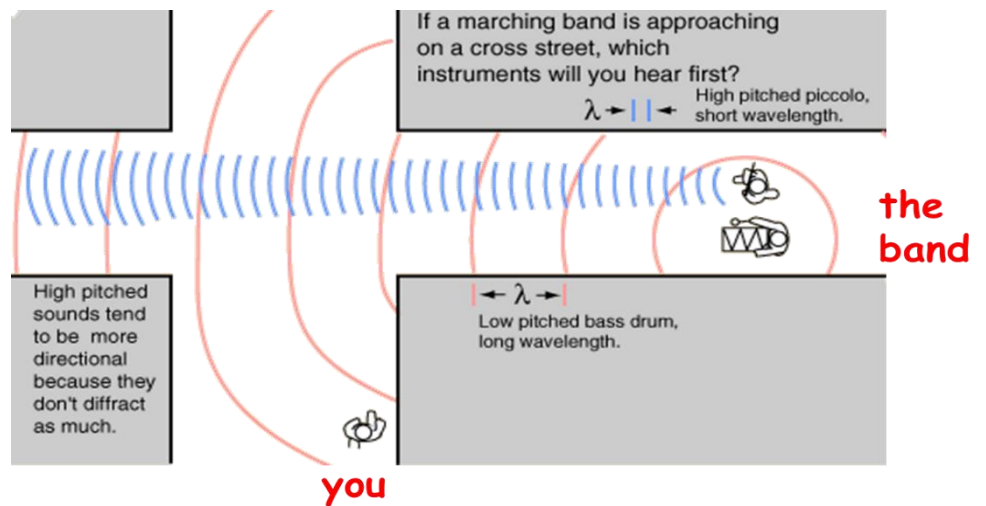
When sound comes across gaps or corners in the medium boundary it gets diffracted.



The amount of diffraction depends on the gap width.
Perfect diffraction occurs when gap width = wavelength.

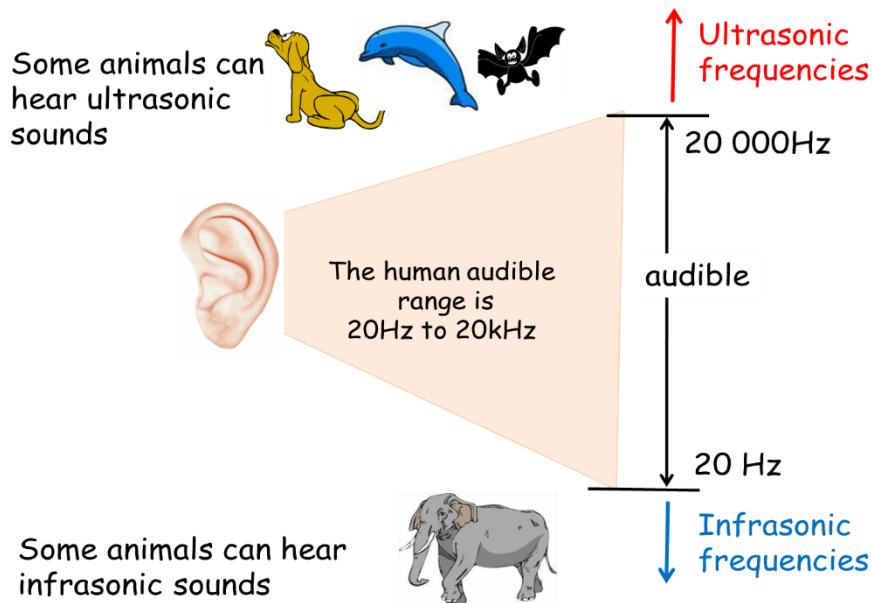
Worked example :

Which instrument will you hear first? Explain.



Ultrasound

Definition : Sound waves at ultrasonic frequencies (above 20kHz)



Some Uses of Ultrasound

Ultrasonic cleaning of jewels, tools, instruments. Etc...

The instrument/jewellery to be cleaned is placed in liquid. Ultrasound is transmitted through the liquid. The liquid particles, also vibrating at ultrasonic frequency, clean the instrument surface, including hard to reach places.



Dog Whistles

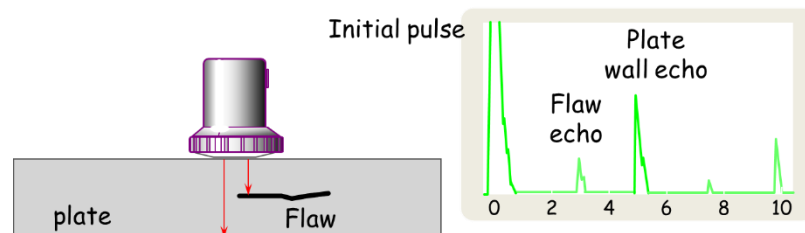
Since dogs can hear ultrasonic waves, we can communicate with dogs without disturbing other people.



NDT (Non Destructive Testing)

This system uses ultrasounds to detect flaws or changes in a piece of machinery, pipe or material etc.

Ultrasound reflecting off a crack or defect sounds different from that reflecting off the rest of the material. It also produces a different signal on the oscilloscope.



Ultrasound Medical Imaging

An ultrasound device transmits the ultrasounds when pressed against the body and then records the echoing waves.

When an ultrasound wave strikes an organ or tissue, it echoes.

By measuring these echo waves it is possible to determine how far away the object is and its size, shape, and consistency (whether the object is solid, filled with fluid, or both).

Sound reflects more strongly from the outside of an organ rather than from its inside. These differences in sound enable the machine to calculate the distance of the particular point of reflection.



With multiple sound transmissions, a real-time image is formed.

Why is ultrasound imaging safer than X-ray imaging?

X-rays are harmful electromagnetic waves.

Ultrasound waves are harmless compared to X-rays

Why is ultrasound better than X-rays for studying internal organs?

Ultrasound can distinguish between different layers of soft tissue while X-rays cannot

Why are X-rays still in use then?

X-rays are better for studying bone because they can penetrate bone, while ultrasound has limited penetration

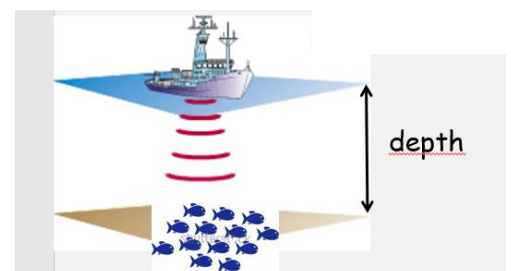
Echolocation by Ultrasound

Used by dolphins, bats, rats and other animals. Also used in SONAR.

For example, dolphins can emit a high-pitched series of clicks in a narrow beam directly in front of them. As the clicking sounds strike an object, such as a fish, some of the sound is reflected back as an echo to the dolphin, who can hear the sounds and perceive a image of the object or fish.



Echosounders located on the ship send signals and then listen for the returning echoes. The time taken for the echoes to return is recorded. The system would already know the speed of sound in that particular area (which depends on temperature, pressure, salinity, etc).



Worked example:

A fishing trawler out at sea is scanning the ocean bed in search of a large shoal of tuna. The trawler sends a sound wave of wavelength of 1.5mm down to the ocean bed. The frequency of the waves is 1MHz. The waves return 5seconds later.

a) Is the fishing trawler sending ultrasonic waves? Explain.

--

b) Calculate the speed of these waves in the ocean.

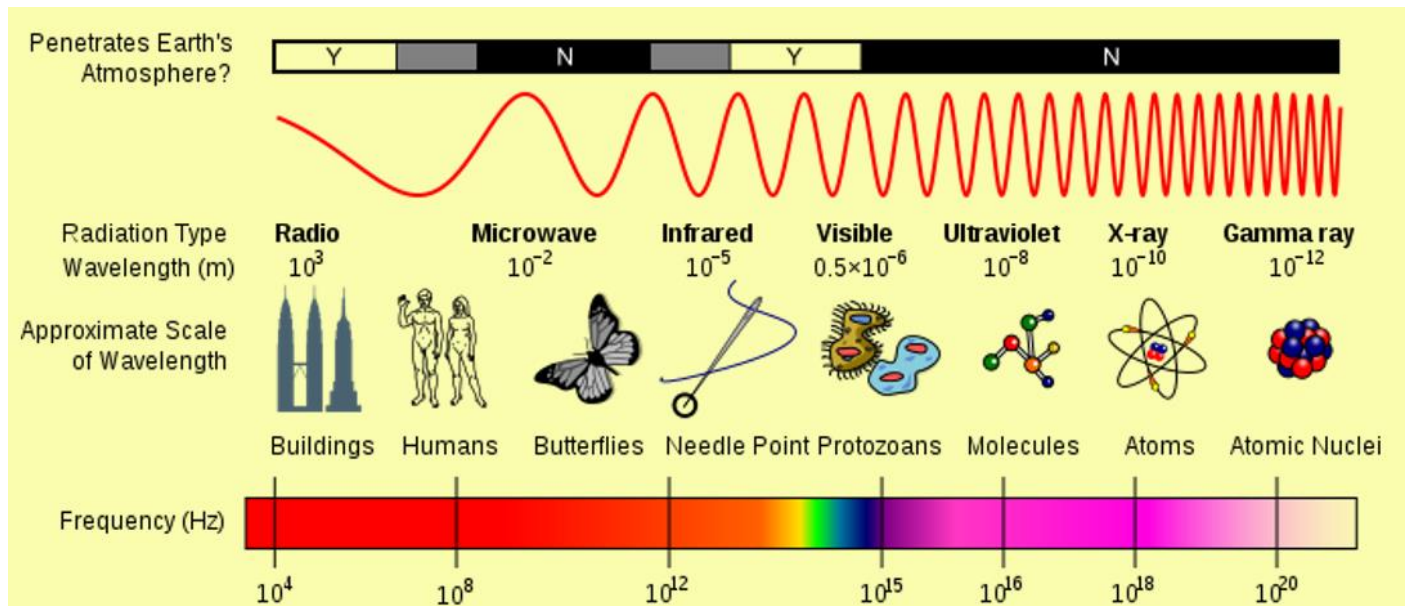
c) Calculate the depth of the ocean bed.

A few minutes later the fishing trawler, still anchored at the same location, sends more waves to the ocean bed. The trawler received two echoes, a strong one 2.5 seconds later, and a weaker one 5seconds later? What do you think this means?

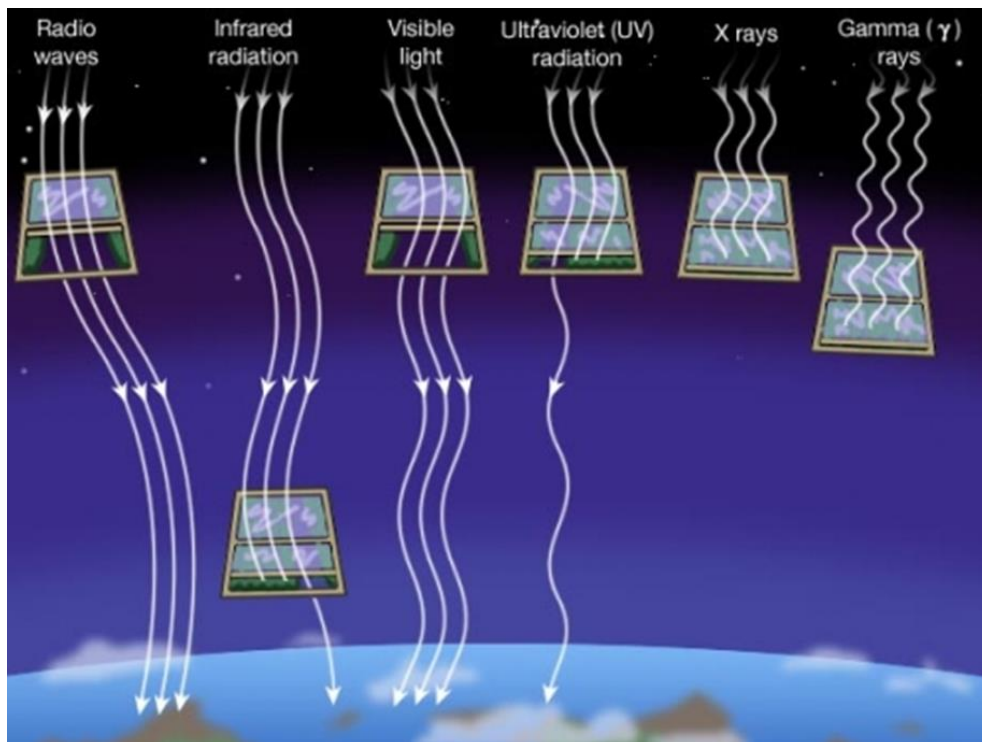
Electromagnetic Waves

What is the Electromagnetic Spectrum?

It is the full range of waves emitted by the Sun



Which of these waves penetrate Earth's atmosphere?

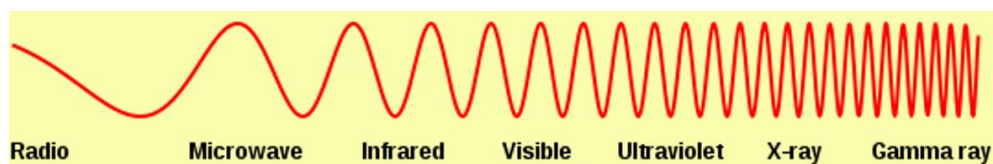


The Three Properties Common to all EM waves

- They are all Transverse waves
- They can all travel through a vacuum
- They all travel at one common speed in a vacuum (3×10^8 m/s)

Differences between the Waves in the EM spectrum

- They all have different wavelengths (therefore different frequencies)



- The shorter the wavelength (consequently the higher the frequency) the more dangerous they are, mainly because they have higher energy levels and can penetrate deeper inside our body.

How to remember the order of waves in the spectrum

Highest frequency
Smallest wavelength

	G ianni	X tara	U mbrella	V jola	I kbar	M inn	R asu
Radiation Type	Gamma Ray	X-ray	Ultraviolet	Visible	Infrared	Microwave	Radio
Wavelength (m)	10^{-12}	10^{-10}	10^{-8}	5×10^{-6}	10^{-5}	10^{-1}	10^3



Electromagnetic wave

Properties & Uses

Gamma

Gamma rays have the smallest wavelength in the spectrum and are very penetrating. Gamma rays are ionizing radiation. They are very dangerous to humans. They are produced by the Sun or by radioactive substances and are detected by specialized equipment. They are used to kill cancer cells and to sterilise food and hospital equipment.

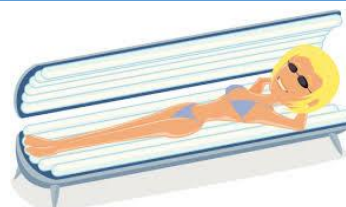
X-rays



X-rays have short wavelengths too. They too are ionizing radiation and are harmful to humans in large amounts. They are produced by the Sun or by X-Ray machines. They too are detected by specialized equipment. Doctors and dentists use X-rays to medical images of bones. X-rays are also used to check baggage in airports and to check for damaged metal parts

Ultraviolet

UV rays also have short wavelengths. They too are ionizing radiation and are harmful to our skin in large amounts (UV index). UV rays are produced by the Sun or UV machines such as tanning beds, halogen or fluorescent lighting (like flood lights). They are detected by special equipment. UV rays have many uses, including tanning beds (since small amounts change the colour of our skin), forged bank notes detection machines and to dry tooth fillings.



They are harmful (UV index).

Visible light

Visible light is the only part of the spectrum that we can see. It is produced by the Sun and other sources of light such as light bulbs and hot objects. Visible light is detected by our eyes or by electronic light sensors. Visible light is used to help us see other objects, in photography and in communication technology (fibre optic cables).



Electromagnetic wave

Properties & Uses

Infra-red

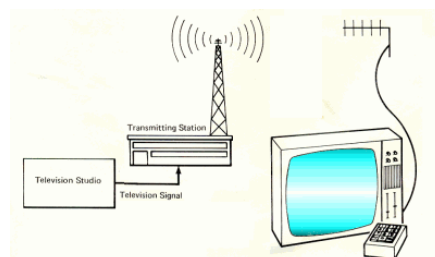
The Sun and matter (at a temperature above 0 Kelvin) emit infra-red rays. IR is detected by our bodies (as heat), by thermometers and electronic heat sensors. Infrared rays are used in remote controls (e.g. TV), thermal imaging (night vision), in burglar alarms, ovens, toasters heaters etc.

Microwaves

Microwaves are Radio waves that have a smaller wavelength than the actual radio waves used for broadcasting. They are produced by the Sun or special magnetic equipment. They are detected by radio receivers (antennas). They are used to heat food in microwave ovens and for satellite communication (mobile phones and GPS).

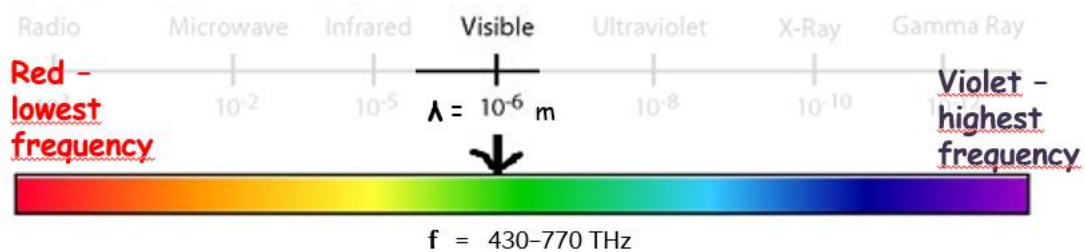
Radio waves

Radio waves have the longest wavelength. They are produced by the Sun and Radio transmitters. They are detected by antennas (receivers). They are used to broadcast TV and radio.



Optics

The Behaviour and Properties of Visible Light



Visible light is part of the electromagnetic spectrum

- transverse wave
- Speed $3 \times 10^8 \text{ m/s}$ in a vacuum
- consists of a whole range of colours each with its own wavelength – together they form white light
- can travel through a vacuum
- can be reflected, refracted and diffracted
- travels in straight lines

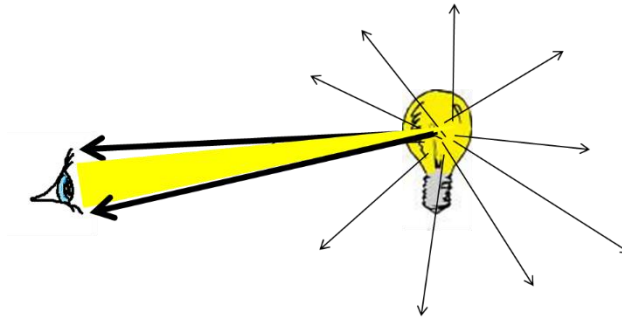
https://phet.colorado.edu/sims/html/color-vision/latest/color-vision_en.html

Sources of Light are LUMINOUS	Reflectors of light are not.
   	   

If we can see something in the light but we cannot see it in the dark, then it reflects light only. Objects that appear black do not reflect any light. They absorb it – black is the absence of light. Most objects reflect some colours (frequencies) and absorb the rest. eg, a green leaf reflects the green frequency and absorbs the rest.

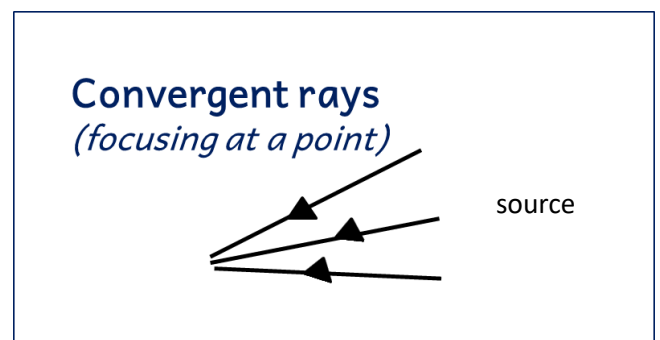
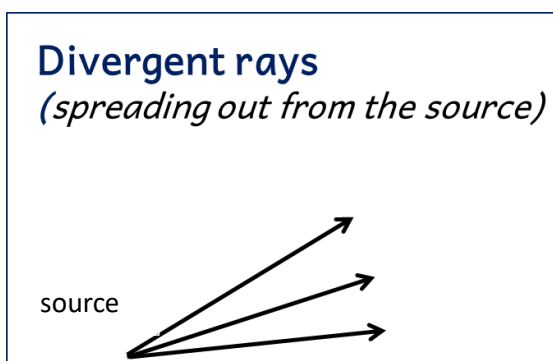
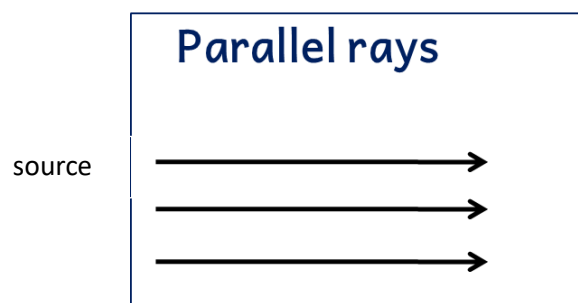
Light Travels in Straight Lines

.....because light takes the shortest path – a straight line



Light cannot move in a curved path...sometimes the rays change direction due to phenomena like refraction and diffraction, giving the impression that the rays are bending, but they still travel in a straight line.

Ray Diagrams to represent Different Wave Directions

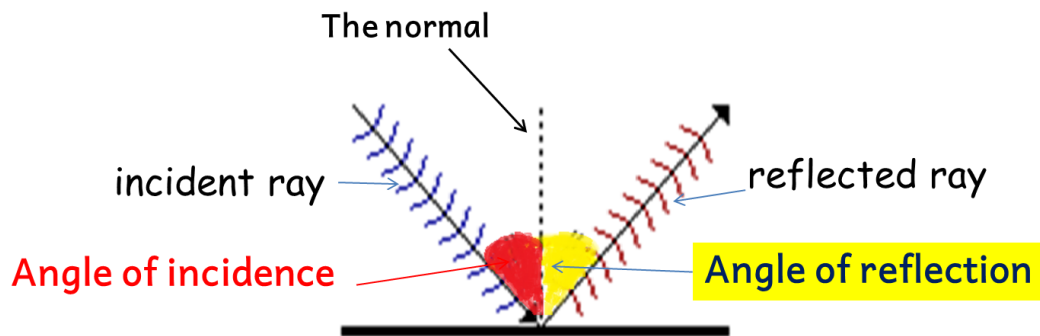


Light Reflection

The Laws of Reflection apply

The light wave bounces off a boundary.

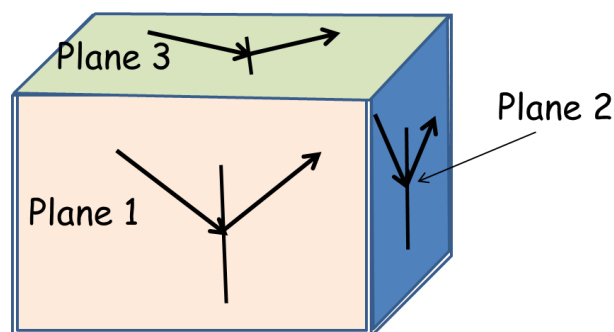
If the surface is smooth enough it may even produce a reflected image that we can see.



1. Angle of incidence = angle of reflection
2. Incident ray, the reflected ray and the normal lie all in the same plane

Explanation of law no.2

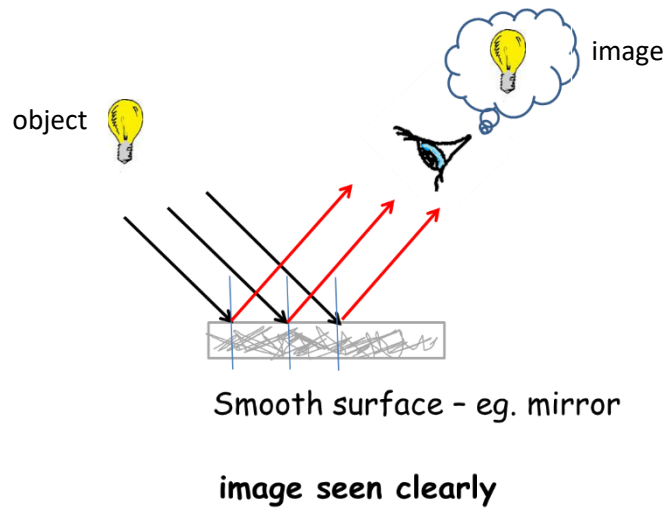
Incident ray, reflected ray and normal lie all in the same plane



This means that incident ray, reflected ray and normal can be drawn on one face of a cube, in 2D only.

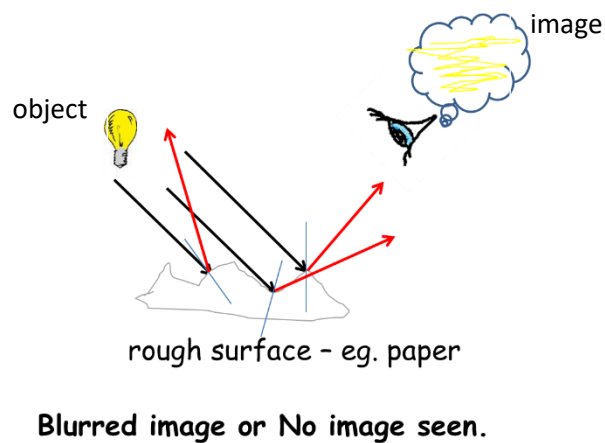
They do not extend to the back or front. This enables us to sketch them on paper, just like a graph.

Specular (or Regular) Reflection



Why is the image seen clearly here?

Diffuse (or Irregular) Reflection

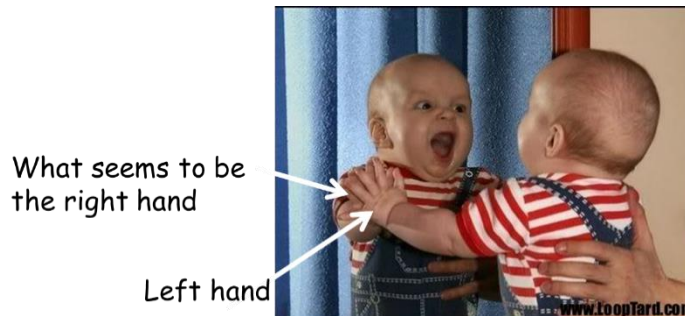


Why is the image seen blurred or none at all here?

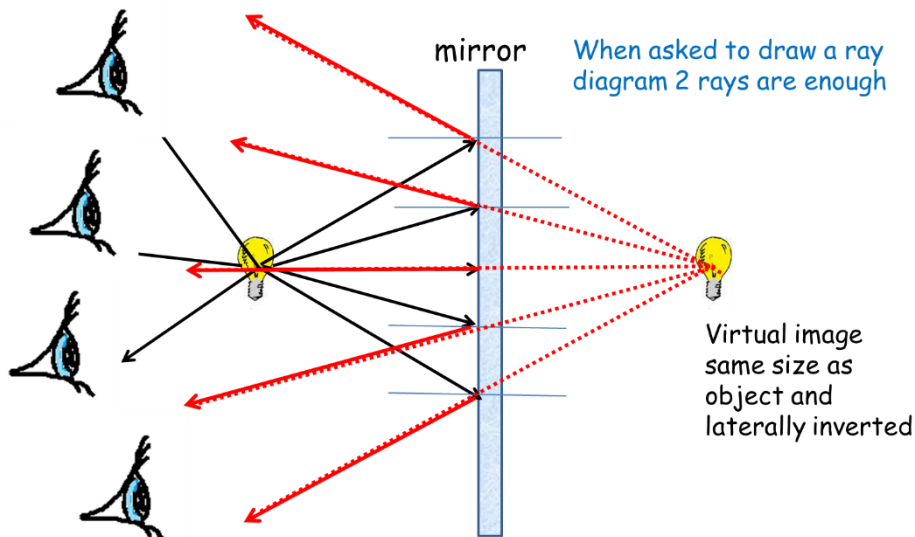
The image reflected by a plane (straight) mirror is **LATERALLY INVERTED**

Laterally inverted means that the left and right side are inverted by the mirror. What is actually on the right of the object is seen as on the left of the image.

The toddler's left hand is touching what seems to be the image's right hand

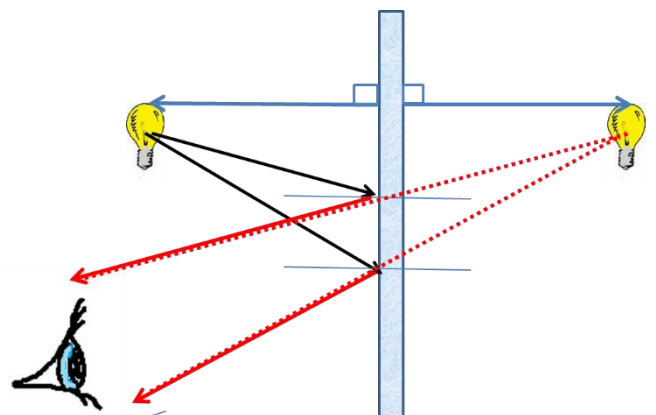


Ray diagram of image formed when an object is reflected from a plane mirror



Important points about the image formed:

1. Virtual image
2. Laterally inverted (back to front)
3. Same size as object
4. Distance from object to mirror = distance from mirror to image
5. Line joining equivalent points on object and image crosses mirror at right angles



The Difference Between a REAL image and a VIRTUAL image

Virtual image – the image seen in a mirror is virtual.

There is no object behind the mirror and no light is actually coming from a point behind the mirror. It cannot be projected on a screen.



Real image – the image projected on the class whiteboard is real.

The projector is converting electrical pulses into coloured rays of light. Every pixel contains many real rays of light that are transferred onto the whiteboard. On the whiteboard you are seeing the original image and not a reflection.



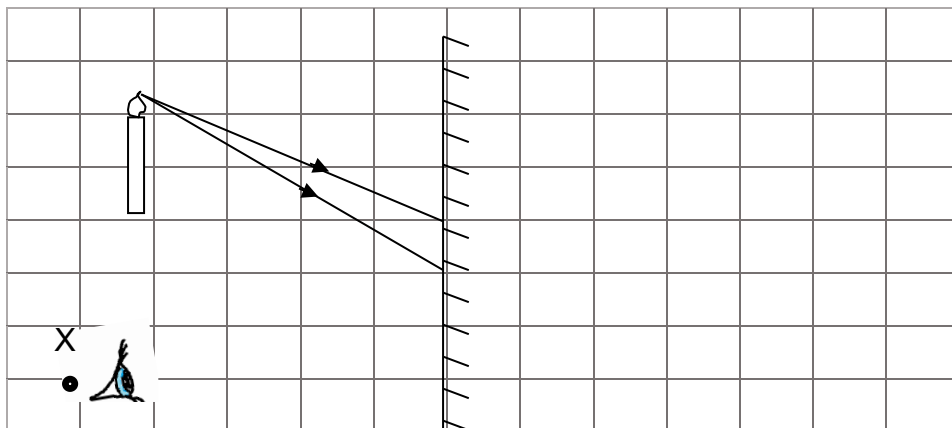
Real image

Visual
studio

Reflection from plane mirrors Worksheet 1

Draw the image seen in the mirror

A burning candle is held in front of a plane mirror as shown below. Light rays from the candle emanate in many directions. Two of the light rays have been drawn.

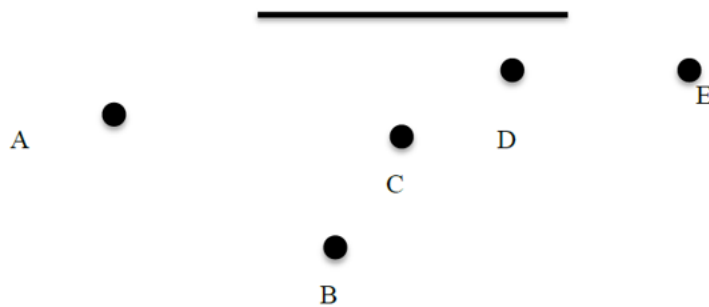


1. What are these light rays called? _____
2. use a ruler and protractor to :
 - a. Draw the normal at the point where each ray meets the mirror.
 - b. Measure the angles of incidence.
 - c. Draw the corresponding reflected rays.
 - d. Draw the image of the candle in its correct position and with its correct size.
3. List four characteristics of the image that you have drawn.
 - i. _____
 - ii. _____
 - iii. _____
 - iv. _____

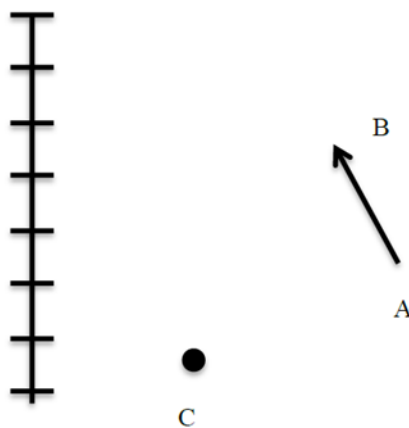
Reflection from plane mirrors Worksheet 2

1. Five students are seated in front of a mirror at positions A, B, C, D and E as shown in the diagram.

Who can student C see? Use ray diagrams to show your answer.



2. Determine the part of the mirror which is used to see the ray AB from point C. You must show your ray diagram. Draw your ray from the center of the circle.



Light Refraction

Is light refracted by every medium? The answer is NO.

Some mediums reflect light (eg most objects)

Some mediums absorb light (eg a black surface)

Some mediums refract light – they let it pass through but change its speed

What type of materials are these? **Transparent materials, eg. Glass, water, perspex**

The Effects of Light Refraction

Visual distortion



Immersed part of ball appears to be larger



Immersed part of pencil appears to be shifted

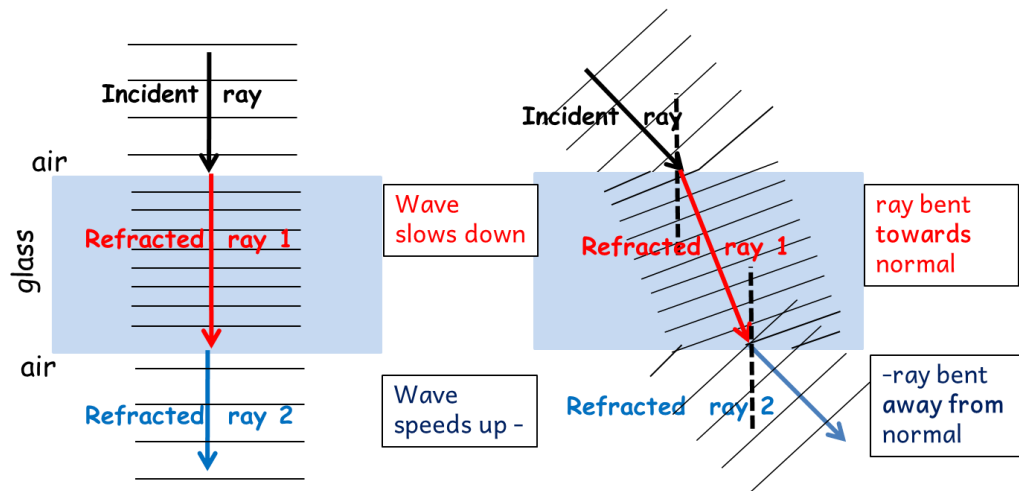


Bottom of glass and ruler appear to be shallower

How can we predict the direction of the refracted ray ?

When light travels into a denser medium (eg from air to glass) its speed decreases (just like deep water to shallow water)

When light travels into a thinner medium (eg from glass to air) its speed increases (just like shallow water to deep water)



What 2 things does the change in speed depend on?



1. The optical density of each medium – represented by a ratio called the refractive index η

2. The angle of incidence at each boundary

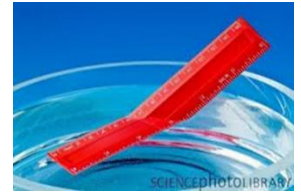
$$\eta_{\text{medium}} = \text{Refractive index of medium} = \frac{\text{Speed of light in air}}{\text{Speed of light in medium}}$$

Refractive index has no unit because it is a ratio

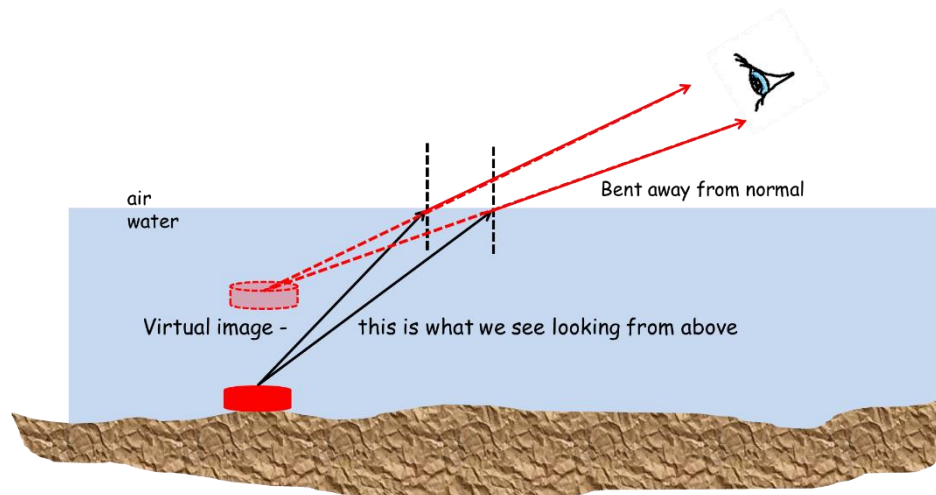
Worked example:

Calculate η for perspex, if the speed of light in perspex is $2.02 \times 10^8 \text{ m/s}$. Take speed of light in air to be $3 \times 10^8 \text{ m/s}$

Visual Distortion effects can be explained by drawing ray diagrams

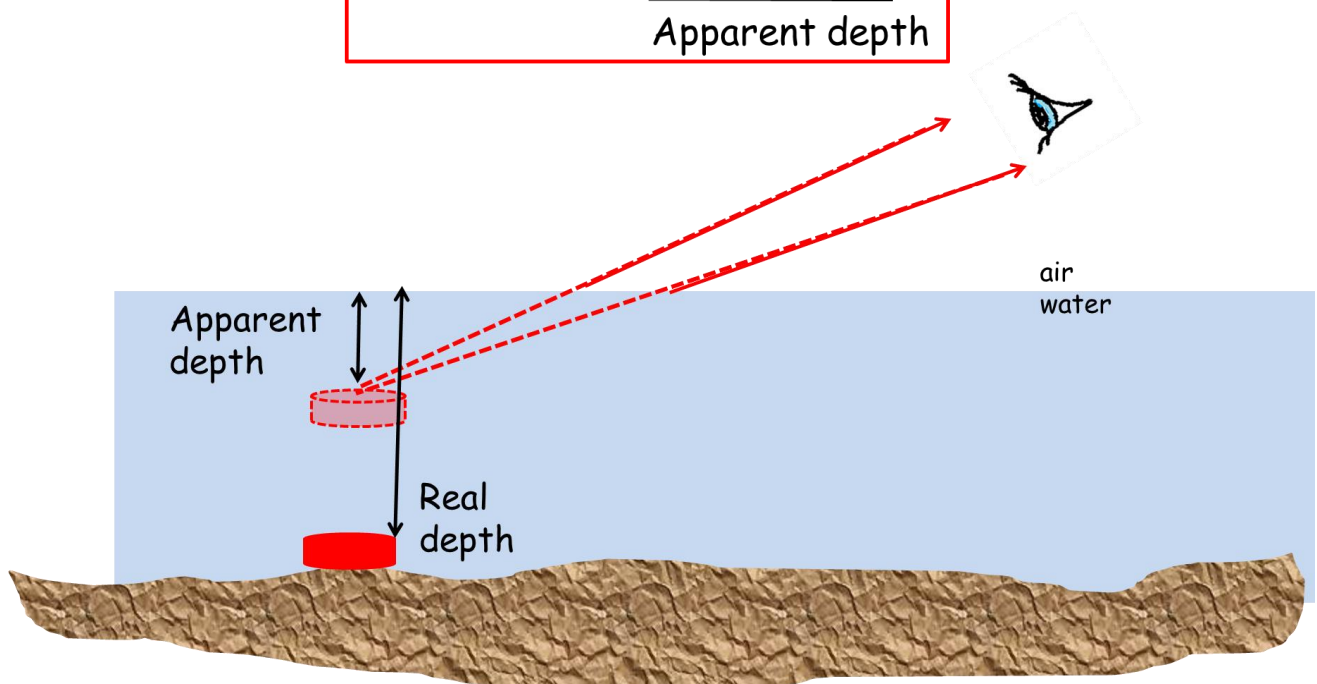


Why do immersed objects appear to be shallower than they really are?



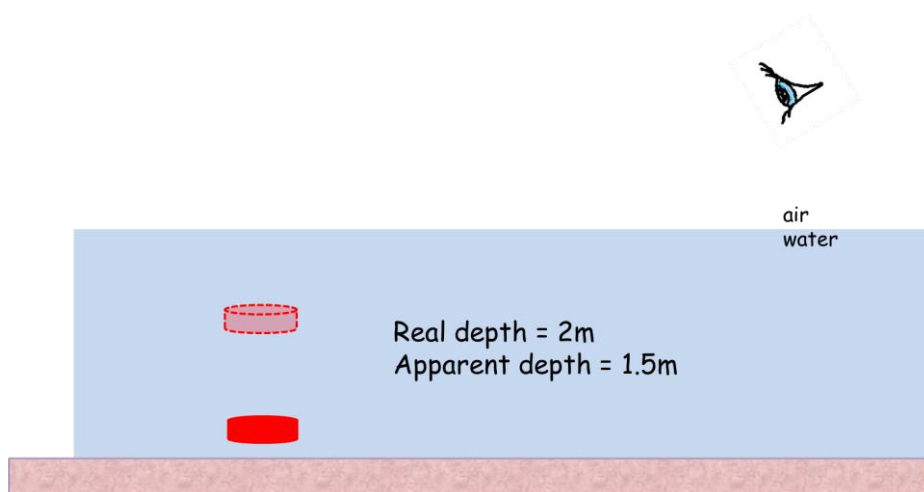
Refractive index of medium : new formula

$$n_{\text{medium}} = \frac{\text{Real depth}}{\text{Apparent depth}}$$



Worked example

The pool is 2m deep but the coin appears to be 1.5m below the surface of the water.
Calculate the refractive index of water.



Refractive Index Worksheet

The refractive index of a medium has symbol n and is given by :

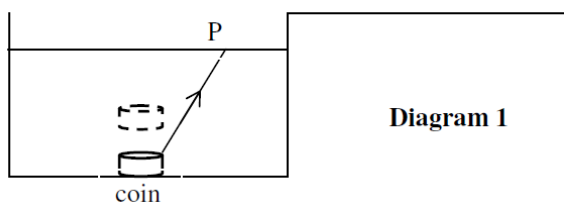
$$n = \frac{\text{speed of light in air}}{\text{speed of light in medium}}$$

or

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$

Task 1: Matsec question (level 5)

Joanne goes to the pool with her family. She spots a shiny coin at the bottom of the pool.



1. What causes the coin to seem so close ? _____
2. Draw the ray of light, as it emerges from P.
3. Draw the normal, the angle of incidence and the angle of refraction as the ray of light emerges from under the water.

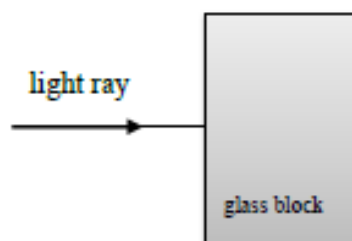
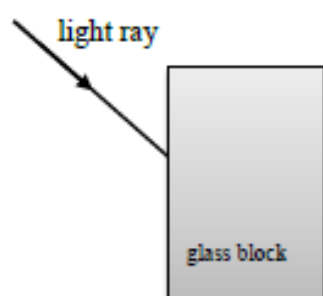
4. The pool is 2.0 m deep but the coin appears to be only 1.5m below the surface of the water. Calculate the refractive index of the water.

5. The speed of light in air is 3×10^8 m/s. Use the refractive index of water to calculate the speed of light in water.

6. Willebrord Snellius was a Dutch astronomer who studied refraction of light waves in the 17th Century.



- a. Complete the following diagrams.



(3 marks)

- b. Name two changes that occur when light is refracted.

- c. A fisherman sees a fish at an apparent depth below the surface of the water of 175 cm. Given that the refractive index of water is 1.33, what is the true depth, in metres, of the fish below the water's surface?

[2 marks]

- d. Swimming pool owners fix depth markers to indicate the depth of the pool. With the aid of a ray diagram, explain why a pool filled with water looks shallower.



air

water

[3 marks]

floor of pool

Total Internal Reflection

The Physics behind Fibre Optics

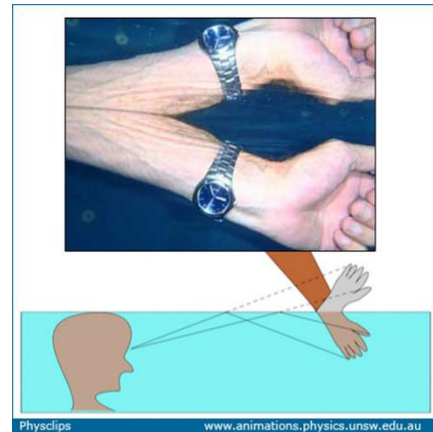


Have you ever looked up at the surface while ducking?

Why does the surface look shiny?

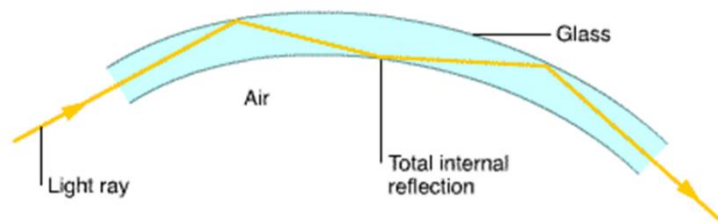
Why can you see the reflection of your hands ?

Why is the surface acting like a mirror? It is transparent!



What is Total Internal Reflection ?

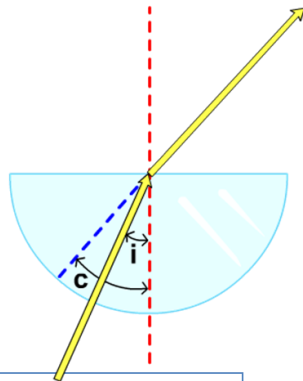
The inside surface of any transparent material can reflect light rather than let it exit the material. The value of the angle of incidence of the exiting ray determines whether the ray will go out or be reflected back inside.



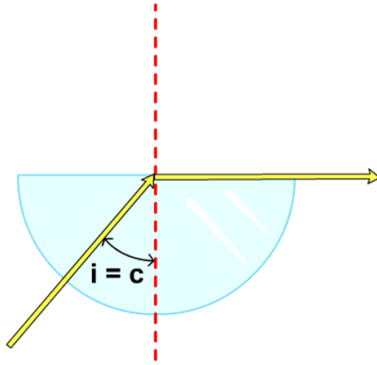
There are two conditions for TIR to occur

1. Light must be travelling from an optically dense medium (high n) to a less optically dense medium (low n), for example, from glass to air, or from water to air.
2. The angle of incidence of the exiting ray must be greater than the critical angle of the medium. For example, if a light ray is exiting from glass to air, the angle of incidence must be greater than the critical angle of glass.

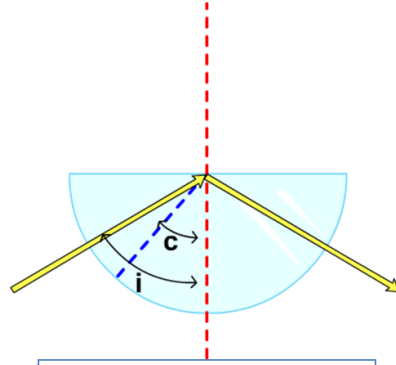
Why is the Angle of Incidence so Important?



If the angle of incidence is **smaller than** the critical angle, then the light ray is refracted.



If the angle of incidence **equals** the critical angle, then the light ray is refracted along the boundary.



If the angle of incidence is **greater than** the critical angle, then the light ray is not refracted and total internal reflection occurs.

The diagram above is part of an experiment to demonstrate TIR.

Note that the shape of the block is semicircular for a specific reason.

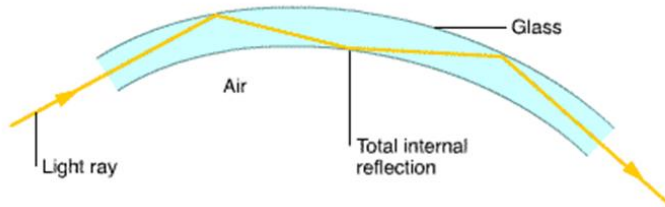
In Mathematics we learn that a line drawn towards the centre of the circle cuts the circumference at 90° . (Radius – tangent theorem – circle theorems)

Applied to Optics, this means that any ray pointed toward the centre of the semicircular block enters at an angle of incidence of 0° , meaning that although the light is refracted, its direction will not change. Only its speed will change. The change in speed alone will not affect the outcome of this experiment, so the semicircular block is ideal in this case.

Uses of Total Internal Reflection

Fibre Optics

Optical fibres are very thin, flexible rods made of special glass or perspex.

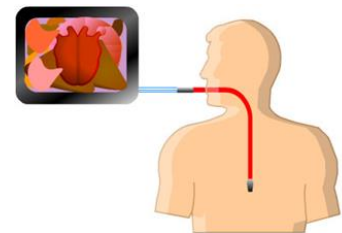


Using total Internal Reflection, light input at the beginning of an optical fibre cable travels inside the cable until it emerges at the other end. There are minimal losses because no light is refracted out of the cable. This is what makes them so good!

Optic fibres are used widely in industry, communications and medicine.

Medicine

Bundles of these fibres can be used to reach inaccessible places such as a patient's stomach, or blood vessels, etc, producing images when connected to appropriate devices.



Machinery testing (Non-destructive testing)

Similarly, optical fibres can reach inaccessible places in machinery that can be tested without breaking them apart.

Communications

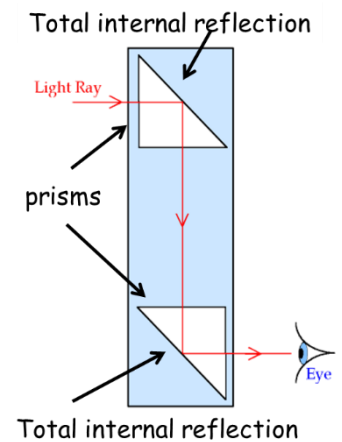
Optical fibres are replacing copper wires and cables in telephone lines, aircraft communications and many other things. You might have optical fibre cables in your own home connected to your TV. Unlike electricity, light is not affected by temperature or changing magnetic fields and so, the signal is clearer.

The Periscope

Periscopes work using prisms or mirrors.

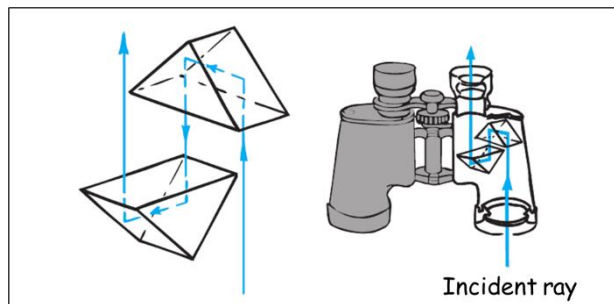


The prisms are placed in such a way that rays of light coming in at the top are incident on the prism wall at an angle that is greater than the critical angle.



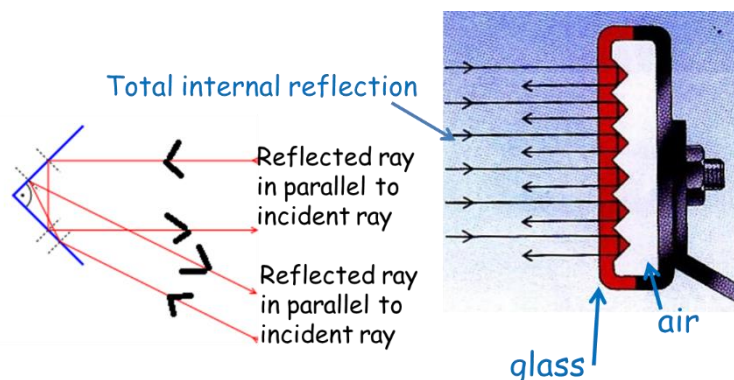
The Binoculars

This also uses prisms that reflect light by total internal reflection. Notice that the binoculars also contains 2 magnifying lenses in each eyehole.



Retro Reflectors

They are intended to avoid accidents by shining brightly towards a driving car.

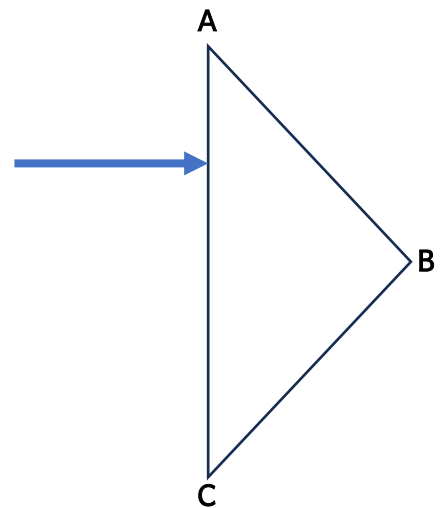


Worked examples

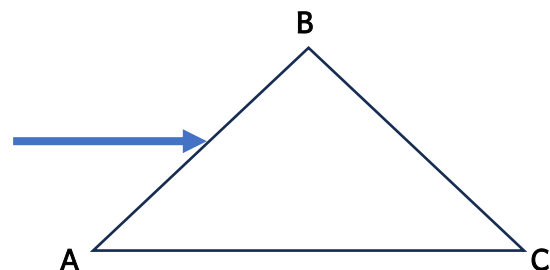
Continue the ray of light as it passes through and exits the prisms.

Triangle ABC is a right angled isosceles triangle, where angle B is 90° .

1. Compare the direction of the exiting ray of light to the original incident ray of light.



2. Compare the direction of the exiting ray of light to the original incident ray of light.



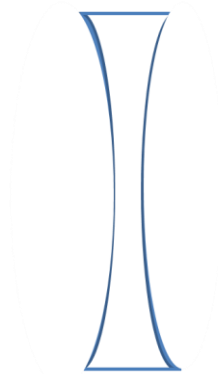
Lenses

There are two basic types of lenses.

Convex (converging) lens
Thickest at the centre

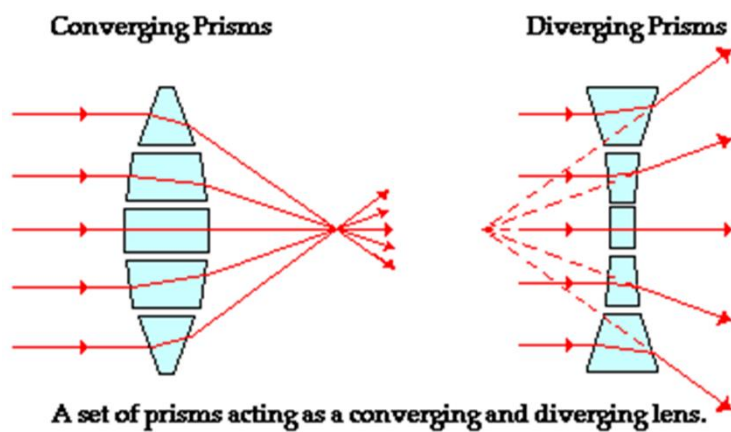


Concave (diverging) lens
Thinnest at the centre



We can obtain information about the images produced by these lenses.

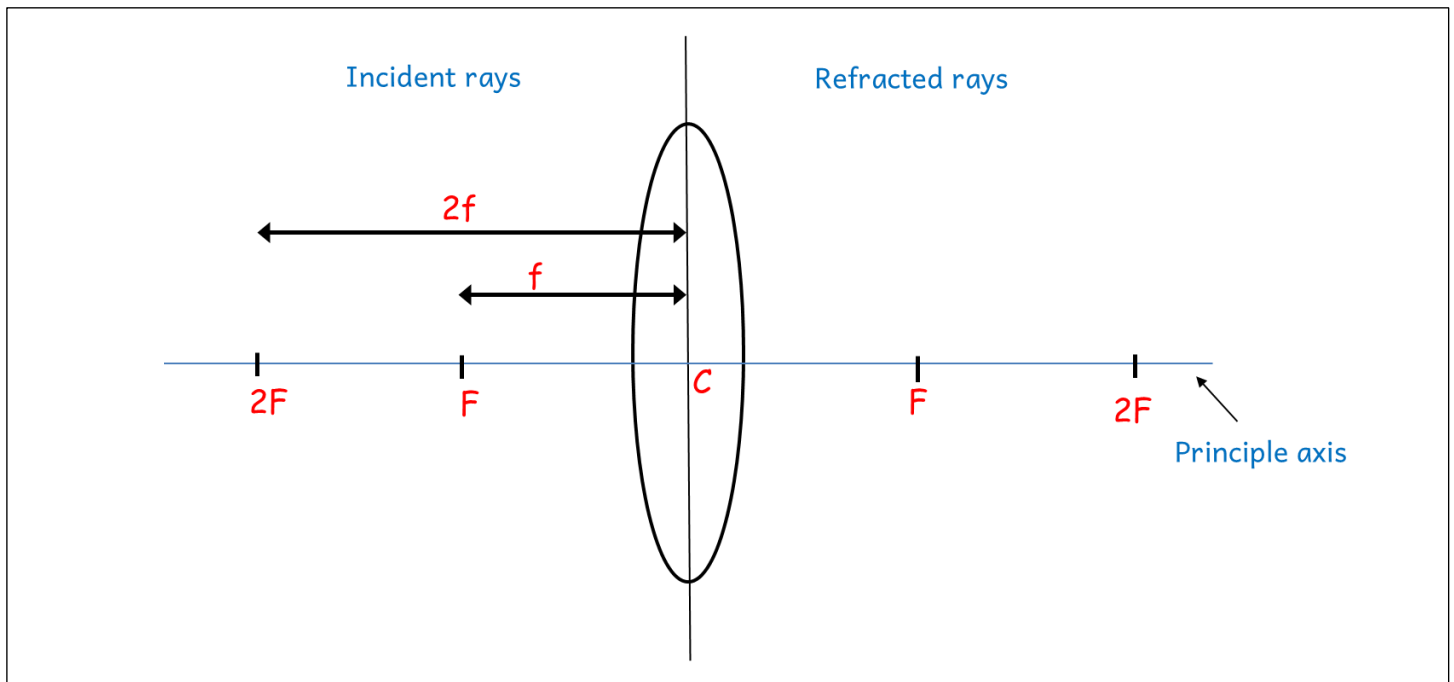
Lenses can be thought of as a series of tiny refracting prisms



Note re. syllabus : We need to be able to draw how rays diverge or converge, but we only need to produce images formed by Converging lenses.

Converging Lenses (Convex)

This is the skeleton axis of lens diagrams. We usually omit the oval and just draw the y-axis to represent the lens.



C – Optical Centre of lens

F – Focus (or focal point) of lens (determined by manufacturer)

2F – 2nd focal point of lens

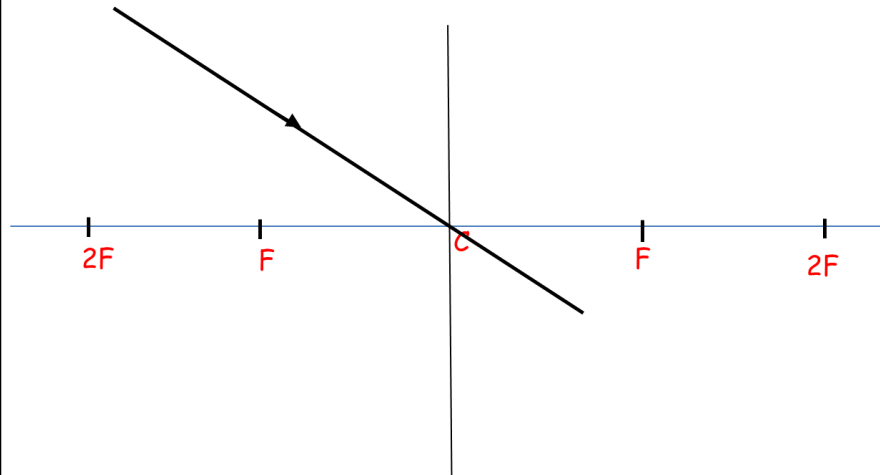
f – focal length of lens – it's the distance between centre of lens and F

It is the point where incoming parallel rays converge to form an image.

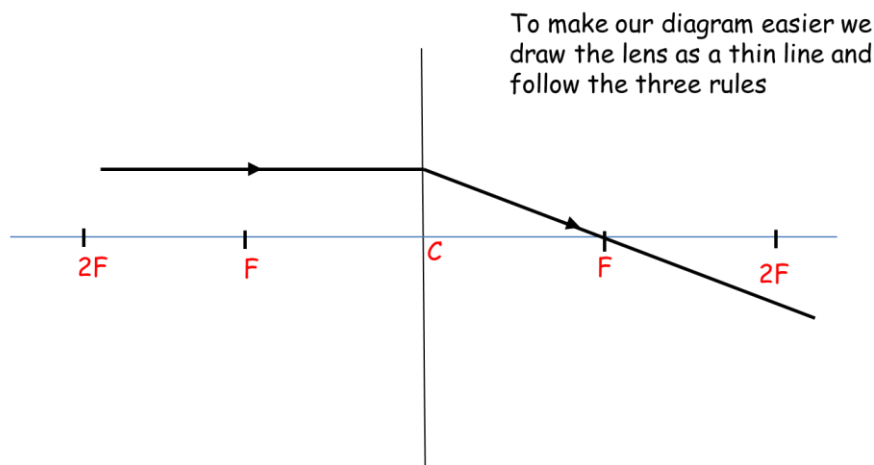
Why are the same points (F, 2F) are marked on both left/right sides of lens?

The Three Rules to Draw Accurate Lens Diagrams

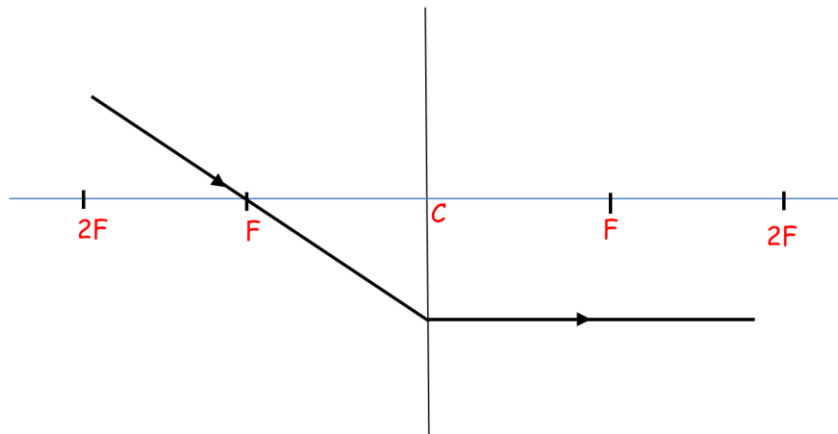
1. An incident ray passing through C is undeviated



2. A incident ray parallel to the principle axis is refracted through F



3. An incident ray passing through F is refracted parallel to the principle axis

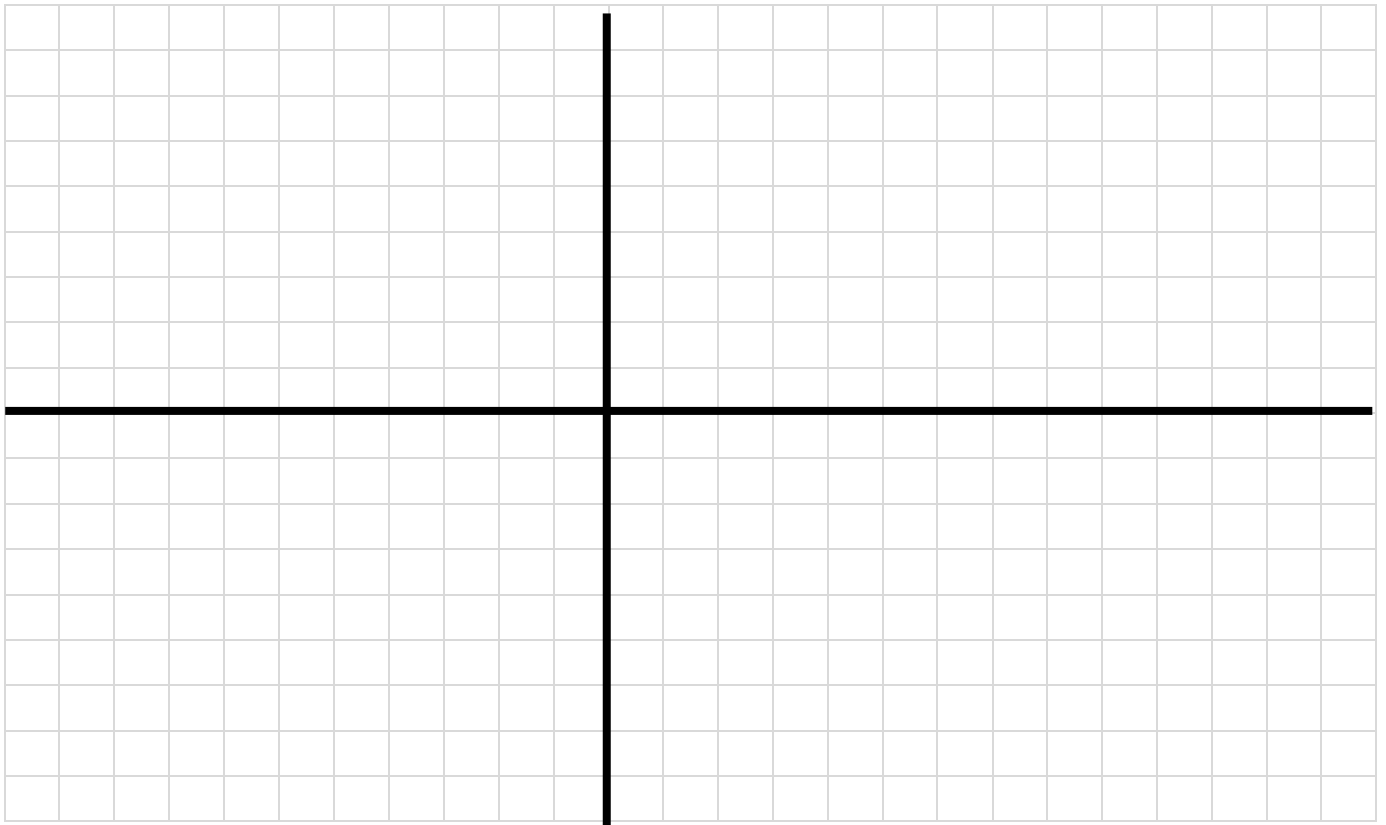


The closer the object moves to the lens, the larger the image

https://phet.colorado.edu/sims/html/geometric-optics/latest/geometric-optics_en.html

There are 6 positions of an object from a convex lens that we need to know how to draw.

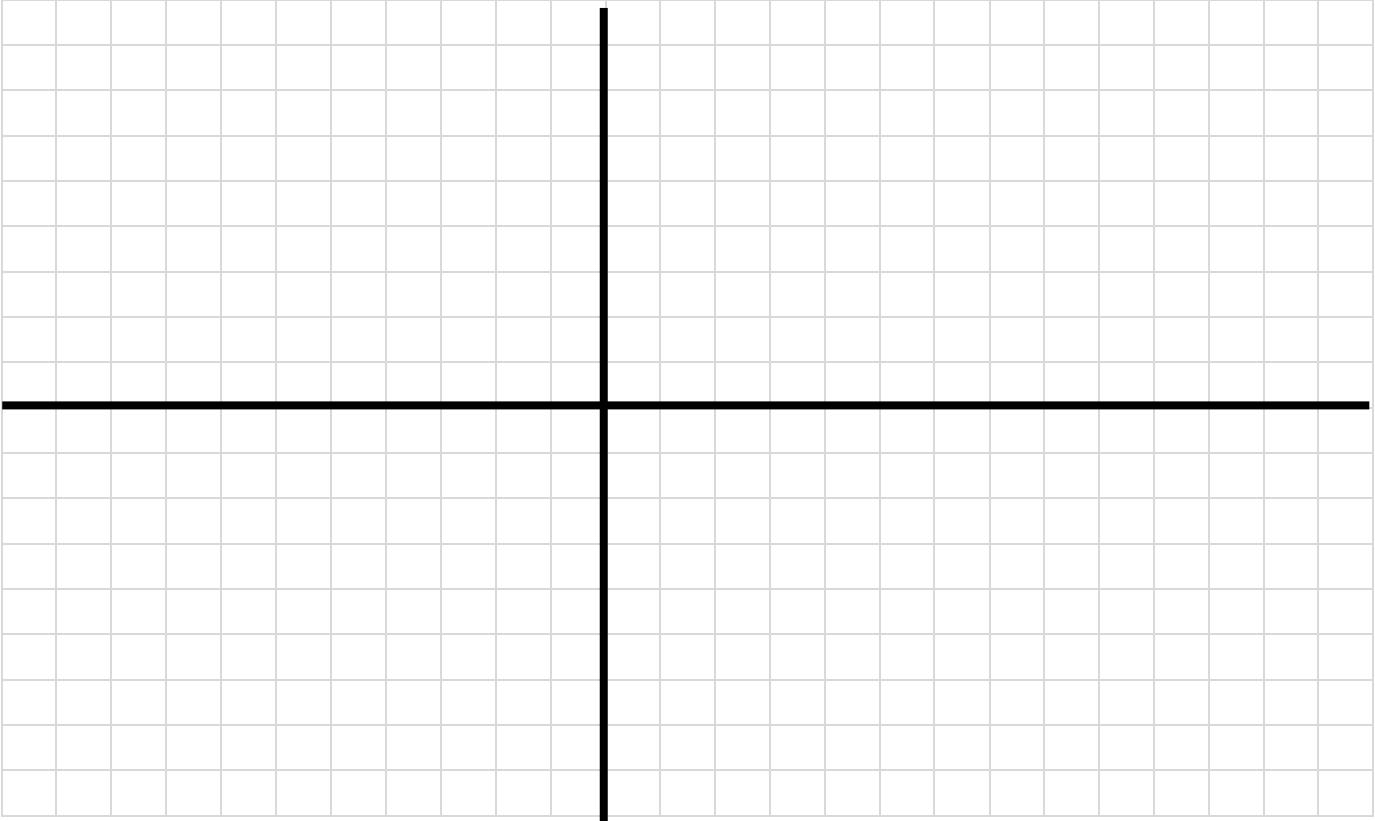
Title of lens diagram 1: _____



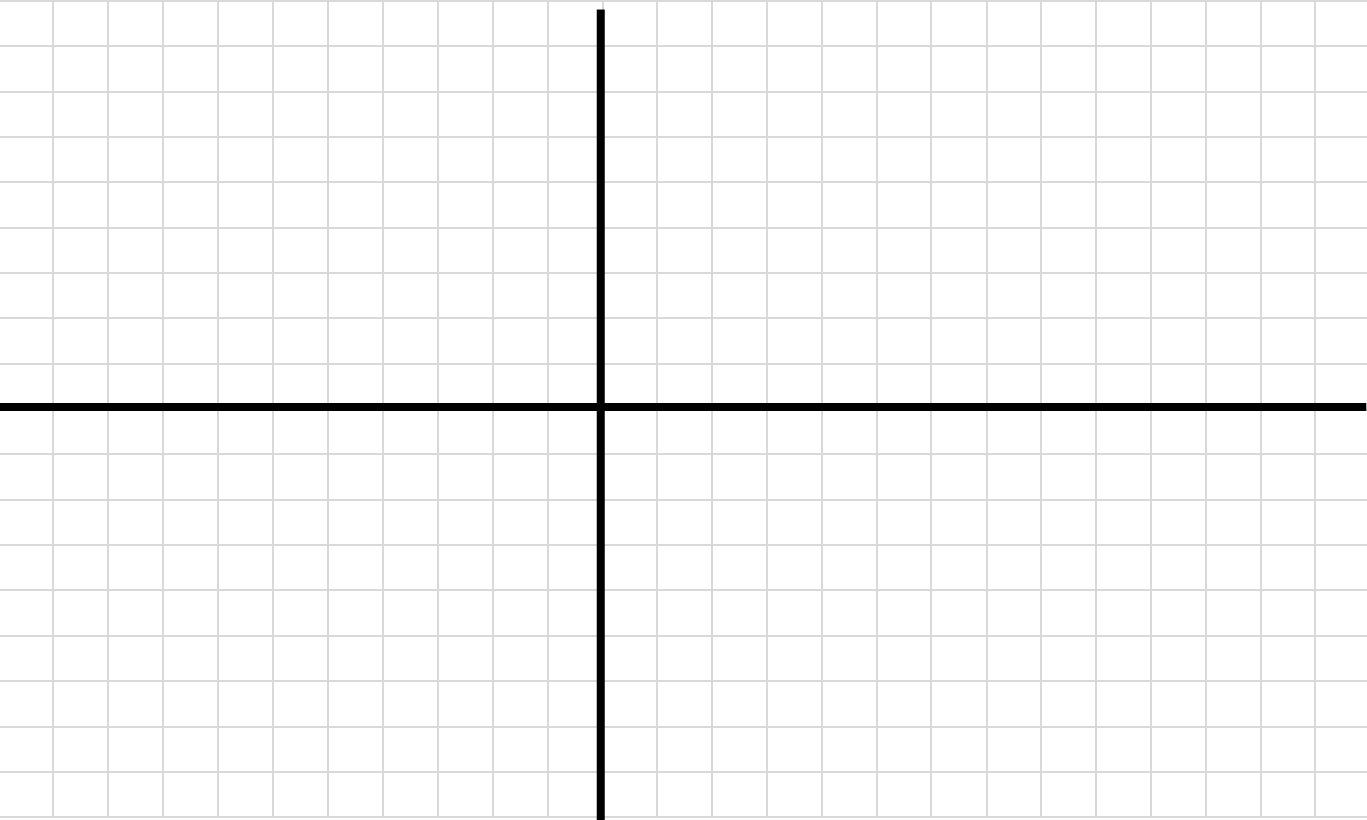
Properties of image formed : _____

Uses of this setup : _____

Title of lens diagram 2: _____


Properties of image formed : _____
Uses of this setup : _____

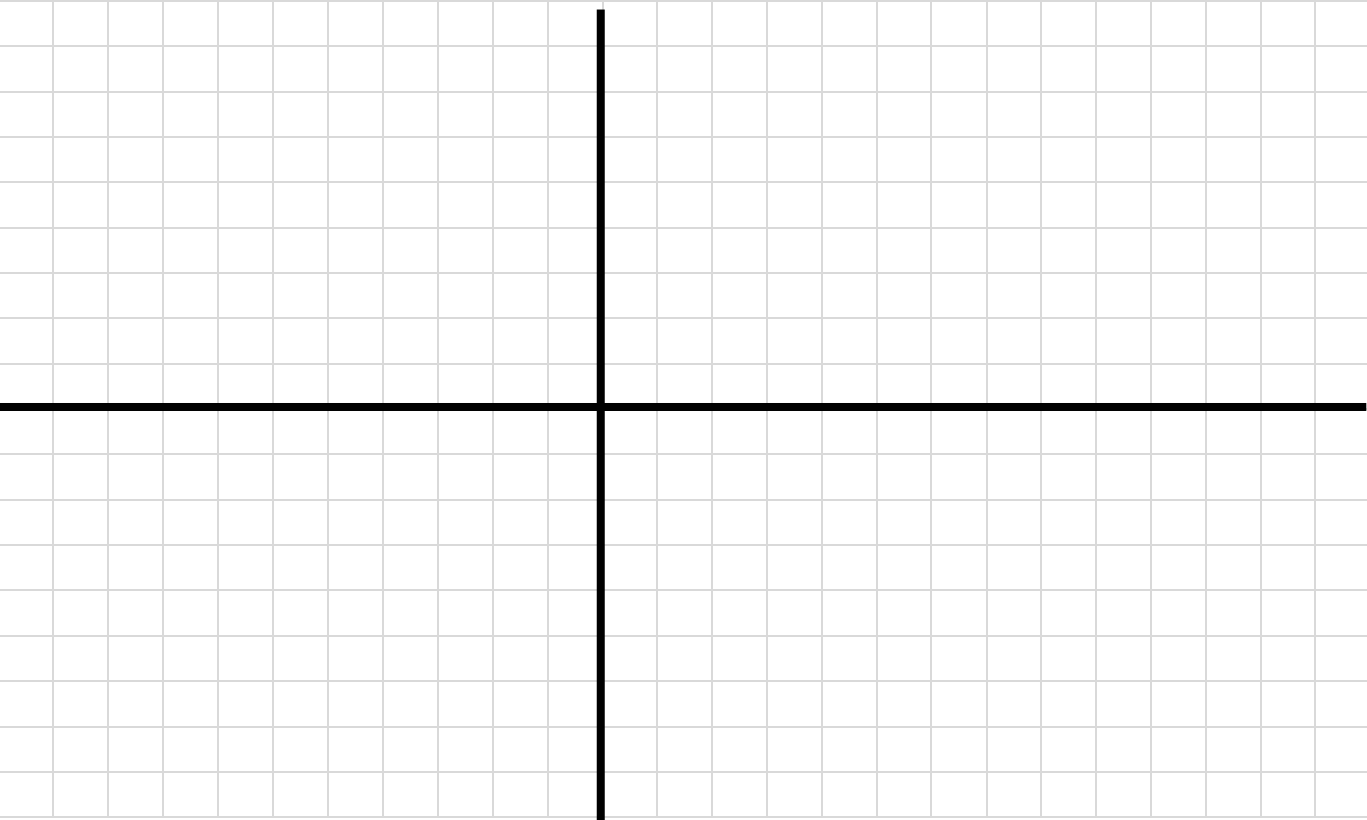
Title of lens diagram 3: _____



Properties of image formed : _____

Uses of this setup : _____

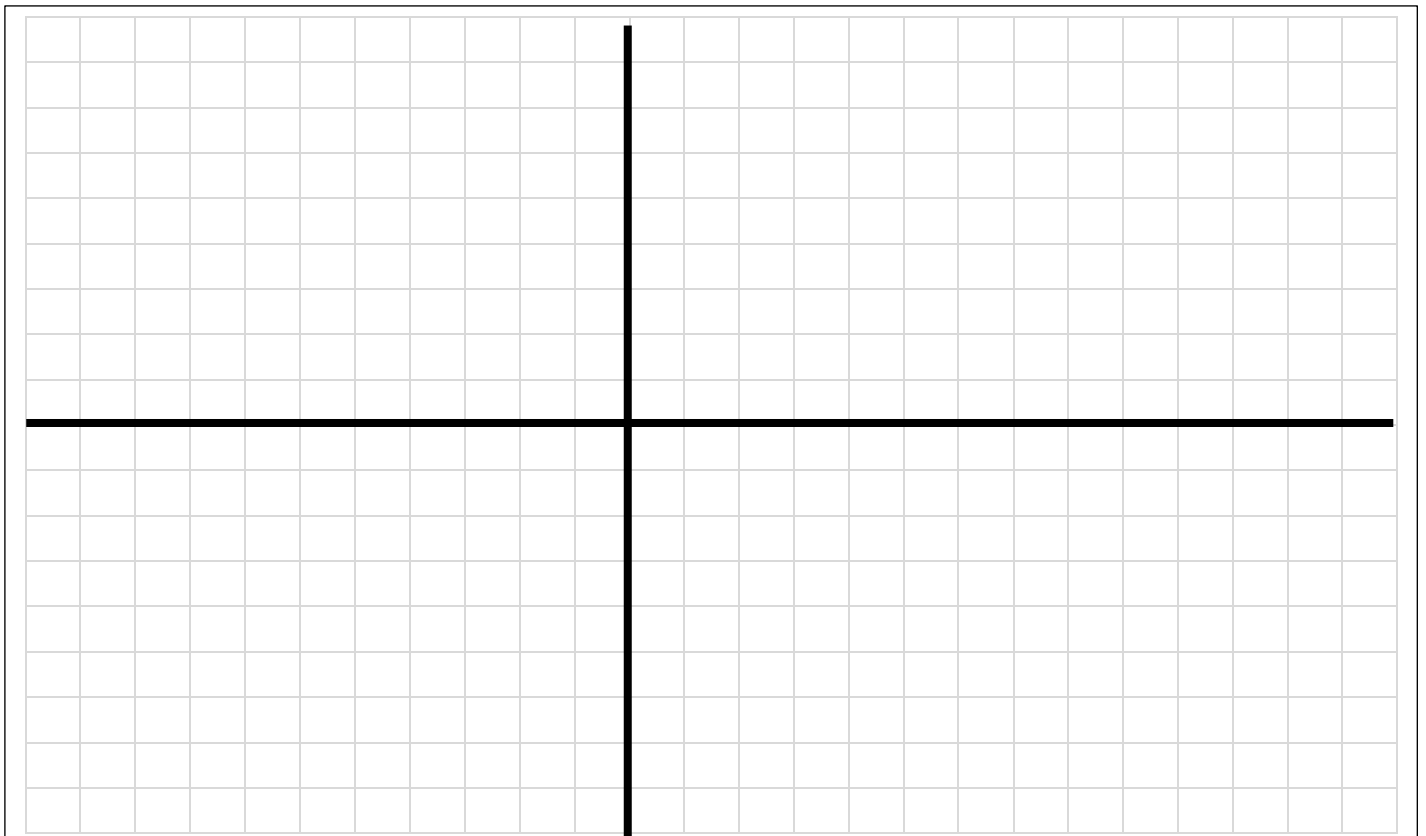
Title of lens diagram 4: _____



Properties of image formed : _____

Uses of this setup : _____

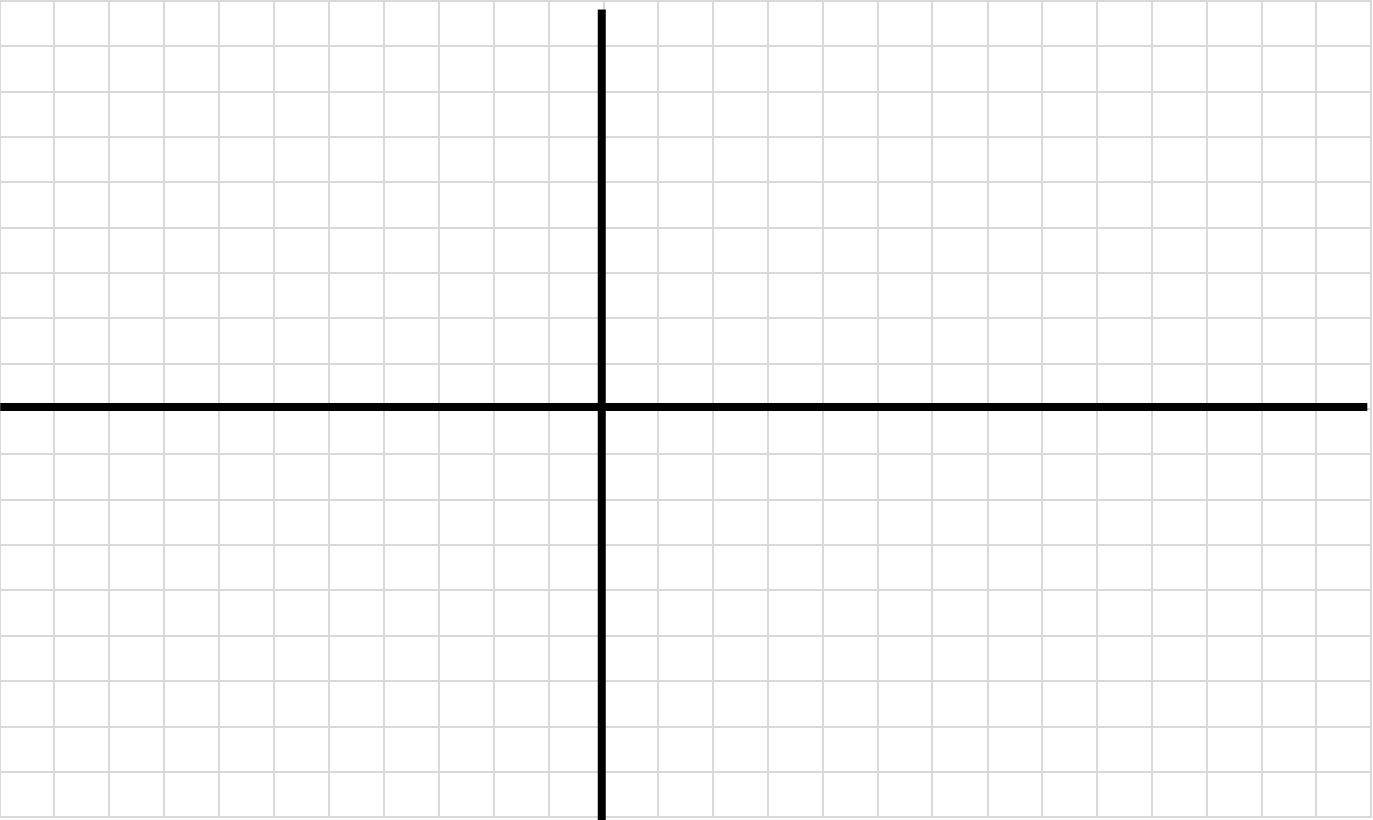
Title of lens diagram 5: _____



Properties of image formed : _____

Uses of this setup : _____

Title of lens diagram 6: _____


Properties of image formed : _____
Uses of this setup : _____

Lens Magnification

A lens can produce images larger than the object or smaller than the object. If the magnification is > 1 then the image is larger than the object, but if the magnification is < 1 then the image is smaller than the object.

Magnification (m) can be calculated using any of two formulae:

$$m = \frac{h_i}{h_o} = \frac{\text{height of image}}{\text{height of object}}$$

OR

$$m = \frac{v}{u} = \frac{\text{distance of image from lens}}{\text{distance of object from lens}}$$

So a magnification of 4 would mean that the height of the image is 4 times bigger than that of the object. It also means that the image is 4 times away from the lens than the object.

NOTE : The same lens can have different values of magnification, depending where the object is placed (see previous ray diagrams). Magnification is not a constant!

Experiment

Focal Length of a Convex Lens – Rough measurement

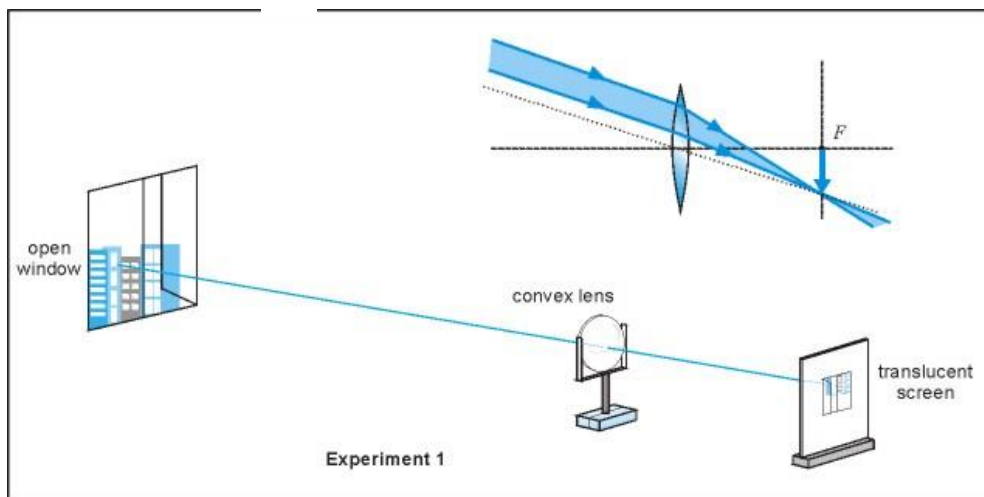
Aim:

To measure the focal length of a convex lens

Apparatus:

Converging Lens, ruler, raybox, mirror, white paper, screen, stands

Diagram:



Method

1. Use the window frame at the far end of the room as a distant object
2. Obtain a sharp image on the screen (as shown in the diagram)
3. Measure the distance from the convex lens to the image (on screen).
4. This distance is the focal length of the convex lens

Precautions:

- Carefully check that the object, lens and screen are perfectly aligned both vertically and horizontally, before starting
- Wait until the sharpest image possible is obtained before taking measurements

Results :

The focal length of the lens is _____

Conclusion:

(You may include an explanation of why you decided to use the above method to measure the focal length)

Experiment

Focal Length of a Convex Lens – accurate measurement

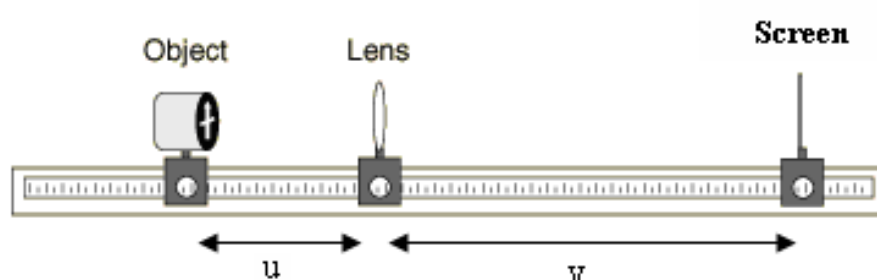
Aim:

To measure the focal length of a convex lens

Apparatus:

Converging Lens, ruler, raybox, mirror, white paper, screen, stands

Diagram:



Note:

u = object distance

v = image distance

Method

1. Set up the apparatus as shown in the diagram above.
2. You should start with the object distance (u) set to 15cm.
3. Move the lens until a sharp image of the object is formed
4. Measure the image distance (v).
5. You should increase the value of u in 3cm steps up to 30cm.
6. For each value of u move the screen to find the focused image and measure the value of v .
7. You will have to calculate the values for $\frac{1}{u}$, $\frac{1}{v}$ and $\frac{1}{u} + \frac{1}{v}$ using a calculator.

Precautions:

- Carefully check that the object, lens and screen are perfectly aligned both vertically and horizontally, before starting
- Wait until the sharpest image possible is obtained before taking measurements

Results :

u (cm)	v (cm)	1/u	1/v	$\frac{1}{u} + \frac{1}{v}$	f (cm)
15					
18					
21					
24					
27					
30					

Calculations:

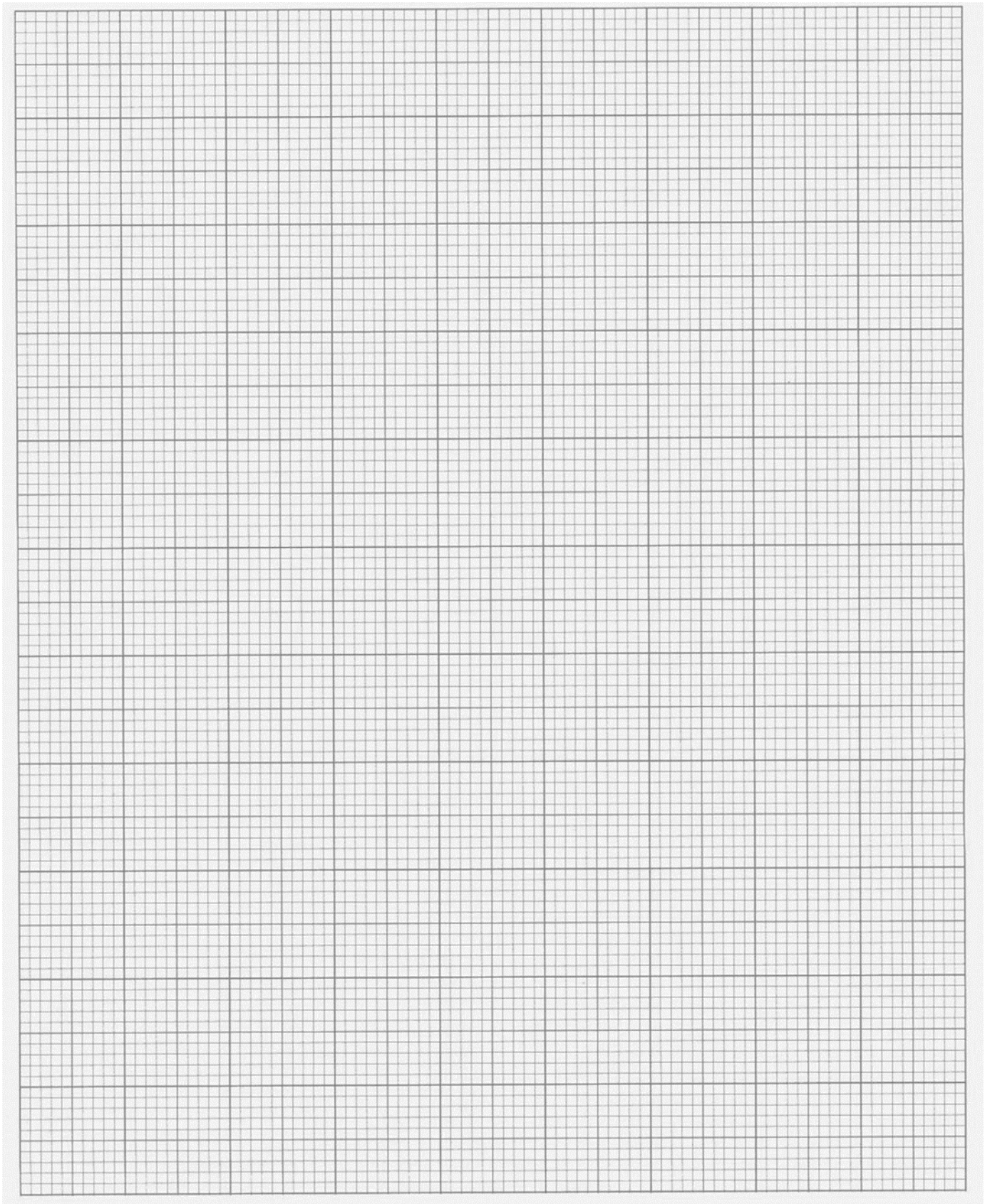
- $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Graph:

- Plot a graph of u (y-axis) against v (x-axis)
- From the graph find the focal length of the lens and compare it to your previous results

Conclusion:

Can you draw any conclusions from your results, calculations and the graph that you have drawn?

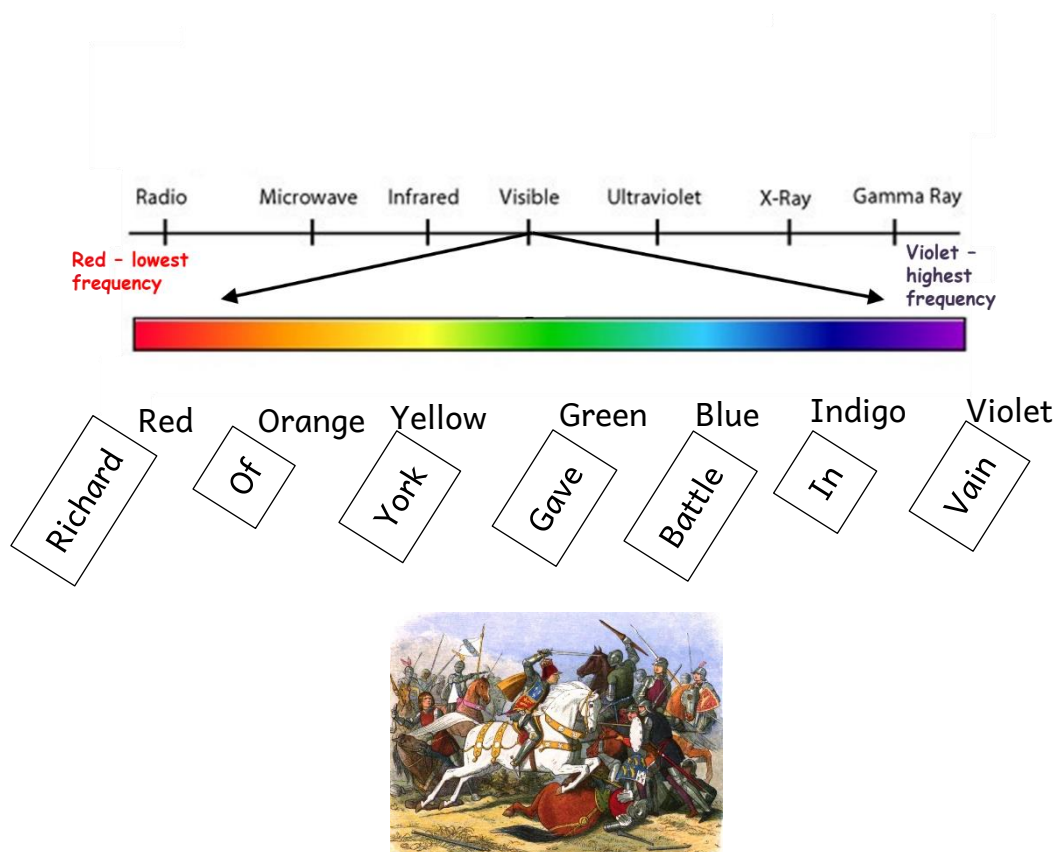


Light Dispersion

The splitting of white light into its component frequencies.



The component frequencies of white light are seen as the seven colours of the rainbow.



Red has the lowest frequency (a little higher than invisible Infra Red radiation) while Violet has the highest frequency (a little lower than invisible Ultra Violet radiation).

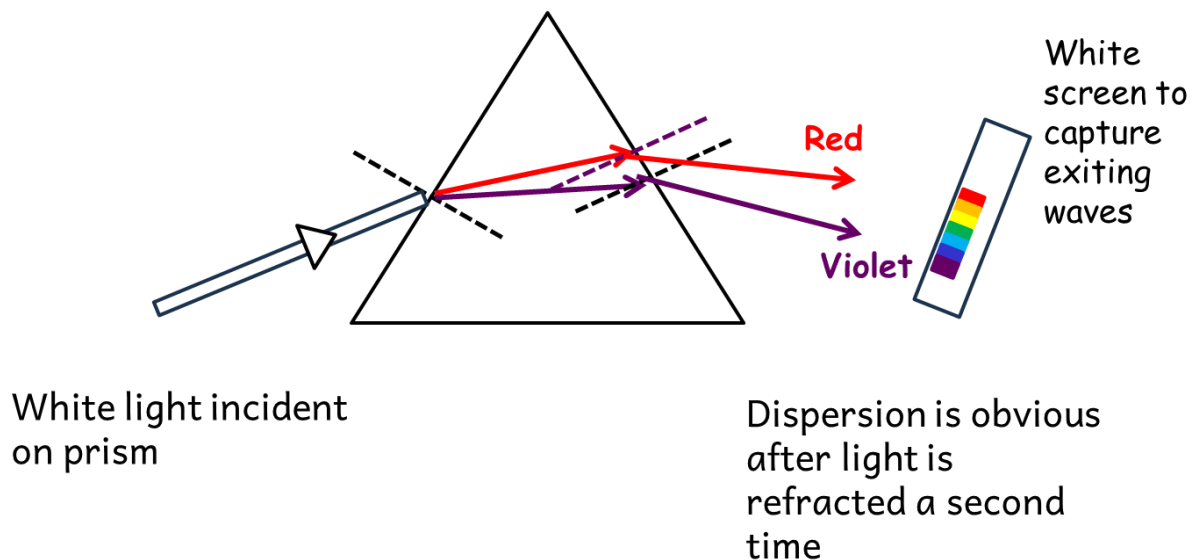
What Causes Light Dispersion?

When white light enters a transparent material at an angle, each of its frequencies is refracted by a different amount.

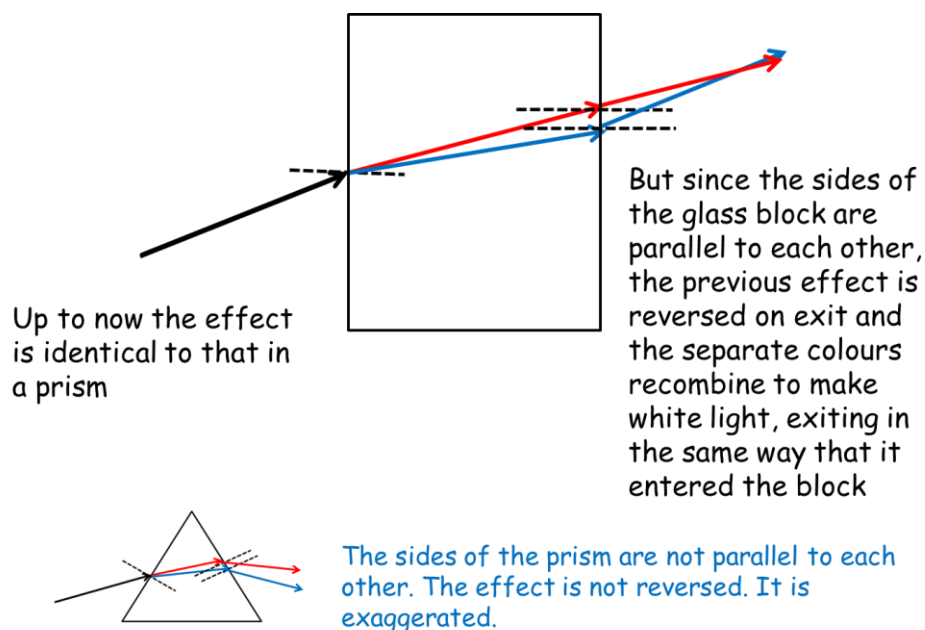
Blue frequencies (higher than red frequencies) are slower than red ones in glass. Therefore they are refracted more.

This effect is exaggerated on exit of a prism.

The final result of double refraction is **dispersion**.



Why does light get dispersed in a prism but not in a glass block?

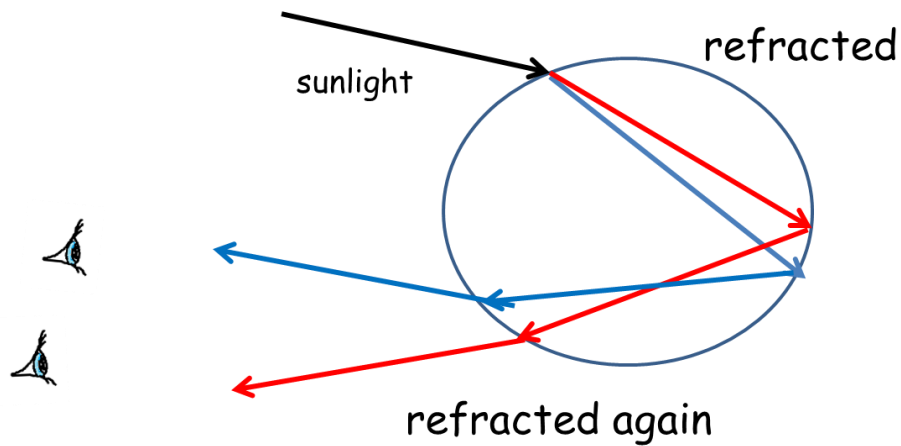


What Causes Rainbows?

For a rainbow to be seen, the sun must be shining while rain is still falling.

Rainbows are caused by the dispersion of sunlight through rain drops.

The colour you see in each drop depends on the angle from which you are observing.



Basic Principles of Waves

1. Name the two different types of waves and explain the differences
.....[1]

2. Define the following and give the S.I. unit of each:

a) Amplitude
.....[1]

b) Wavelength
.....[1]

c) Frequency
.....[1]

d) Periodic time
.....[1]

3. The diagram shows a water wave. 8 waves are produced every 4 seconds

i) Calculate the wavelength and show with a λ on the diagram.
.....[2]

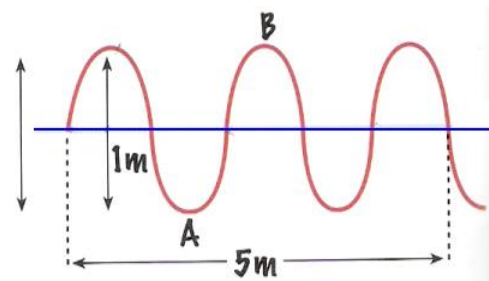
ii) Calculate the frequency
.....[2]

iii) What is the amplitude?
.....[2]

iv) Name the two points A and B

A.....

B.....[1]



4. The diagram shows a wave

a) What type of wave is this?
.....[1]

b) Draw the direction the wave is moving on the diagram, [1]

c) Mark the wavelength on the diagram [1]

d) If the source of the wave is vibrating to and fro 120 times every minute, what is the frequency of the wave?
.....[1]



5. The picture shows Lina making waves with a rope:



a) What type of wave is produced?
.....[1]

b) The frequency of the wave is 2Hz. Explain.
.....[1]

c) Show the amplitude and the wavelength on the diagram

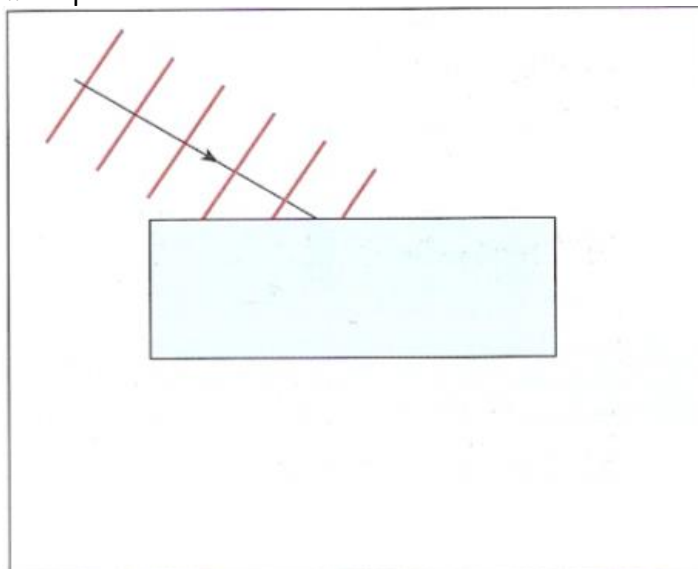
d) The wavelength is 2m. Calculate the speed of the wave
.....[1]

Reflection, Refraction & Diffraction

1. The diagram shows a ripple tank. There is a barrier. Complete the diagram to show reflection of waves. [1]



2. The diagram shows water waves going from deep water to shallow water
a) Continue the path as they go into shallow water and back into the deep water [2]



- b) What is this effect called?

.....[1]

- c) Explain why a wave behaves this way. (In your answer state what happens to frequency, speed, wavelength and amplitude of the wave)

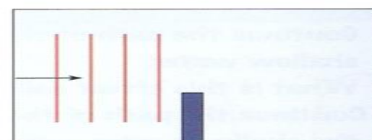
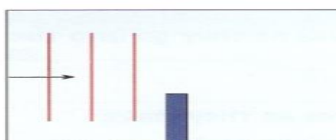
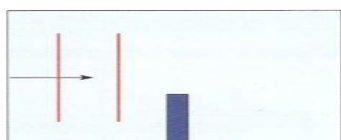
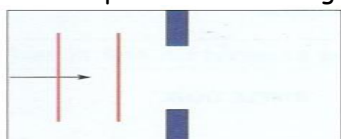
.....

.....

.....

.....[2]

3. Complete the following diagrams [6]



4. ALL Electromagnetic waves travel through air at 300 million m/s

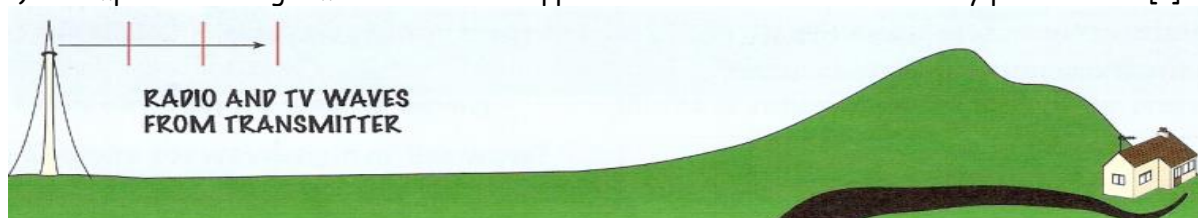
- a) Calculate the wavelength of a radio station that broadcasts at a frequency of 900kHz

.....[1]

- b) Calculate the frequency of a TV station which has waves of wavelength 60cm.

.....[1]

- c) Complete the diagram to show what happens to the radio waves when they pass the hill [1]



- d) What is the name of the effect that occurs on (c)?[1]

- e) Explain why the people in the house would be unlikely to receive TV signals from (b)

.....[1]

Electromagnetic Spectrum

1. The electromagnetic spectrum is a group of waves

--	--	--	--	--	--	--

a) Label A to F with the correct type of electromagnetic wave, starting with the shortest wavelength. [3]

b) Name two common properties of all electromagnetic waves.

.....[1]

2. This table is all mixed up! Match the correct letter (effects on living tissue) and number(uses) with each wave

	Effects on living tissue	Type of wave	Uses
A.	Probably none	<i>Gamma</i>	- Communication - Broadcasting - Radar
B.	Heating of water in tissues can cause burning	<i>X - Ray</i>	Imaging of internal structure in the body
C.	Causes burning of tissue	<i>Ultra Violet</i>	- Fluorescent tubes - Tanning - Checking money
D.	- Kills living cells in high doses - Lower doses can cause cells to become cancerous - Causes tanning	<i>Visible Light</i>	- Kill bacteria in food - Sterilise medical equipment - Treat tumours
E.	- Kills living cells in high doses - Lower doses can cause cells to become cancerous	<i>Infra Red</i>	- Satellite communication - Cooking
F.	- Kills living cells in high doses - Lower doses can cause cells to become cancerous - Kills cancer cells	<i>Microwave</i>	- Radiant heaters - Grills - Remote controls - Night vision
G.	Activates sensitive cells in the retina	<i>Radio</i>	- Seeing - Optical fibres

3. X-rays are part of the EM spectrum.

a) Explain the properties of x-rays which make them useful for us

.....[1]

b) Explain why x-rays can be dangerous to human beings

.....[1]

c) Microwaves are also part of the EM spectrum

i) Give one difference between x-rays and microwaves

.....[1]

ii) Name two uses for microwaves

.....[1]

iii) Recently there has been a lot of concern about the use of mobile phones and its effect on humans. Explain why microwaves could be harmful

Pitch & Loudness

1. The picture shows 'Cikka the Rocker' on her world tour



a) What type of wave is a sound wave?

.....[1]

b) Describe how sound travels from her loudspeaker to the audience

.....

[2]

c) During the sound check the sound engineer recorded two notes from her guitar. The traces from the CRO are shown in A and B.

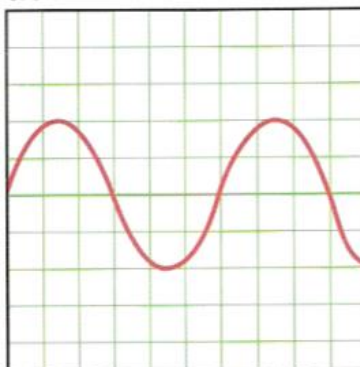
i) Which note is the **loudest**? Explain.

.....
[2]

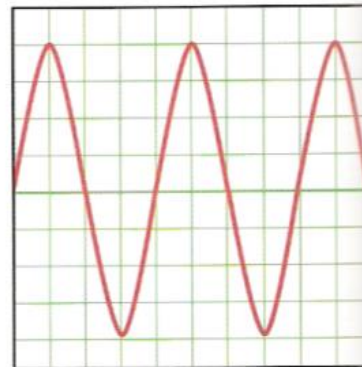
ii) Which has the highest pitch? Explain

.....
[2]

Trace A



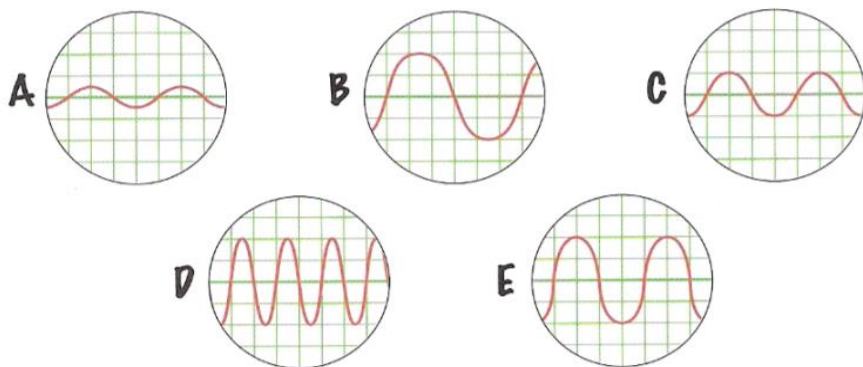
Trace B



d) Cikka then played a note whose loudness AND pitch were doubled compared to the note in trace A. Sketch the trace of this note on A [2]

e) Cikka then played a note whose loudness is a quarter and pitch a half of the note in trace B. Sketch the trace of this note on B. [2]

2. The diagram below shows traces on an oscilloscope screen:



Which show:

- a) The quietest sound
- b) The highest pitch
- c) The loudest sound
- d) The lowest pitch

[4]

3. Choose the correct words:

Increasing the (**frequency / amplitude**) of a sound will (**raise / lower**) the pitch of the sound, producing a (**higher / lower**) note. Decreasing the (**frequency / amplitude**) will (**raise / lower**) the pitch of the sound. Increasing the amplitude will increase the (**pitch / loudness**) of the sound and decreasing the amplitude will make the sound (**louder / quieter / higher / lower**).

Ultrasound

1. Complete the following:

Sounds above 20,000Hz have too high a to be heard by the human ear. Sounds above this frequency are called [2]

2. Pregnant women have ultrasound scans to check on the progress of an unborn baby

a) What is ultrasound?

.....[1]

b) The image produced by ultrasound is not as clear as x-rays, so why is ultrasound used for looking at unborn babies rather than x-rays?

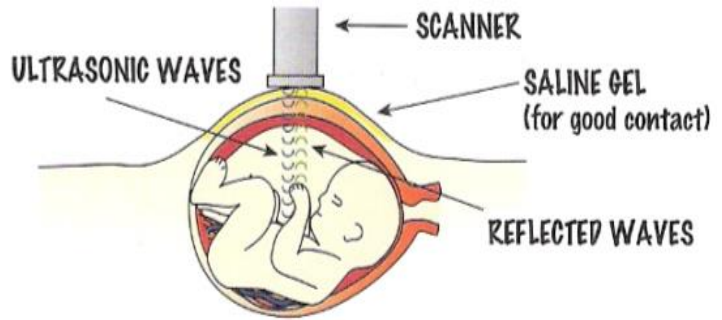
.....

.....[1]

c) Describe how ultrasound is used to produce an image of the baby

.....

.....[1]



3. Eco-sounders can be used to locate submarines

Using the diagram explain how eco-sounders are used to locate sub marines

.....

.....

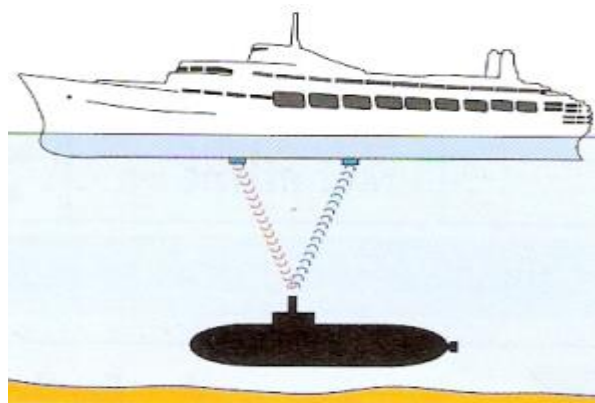
.....

.....

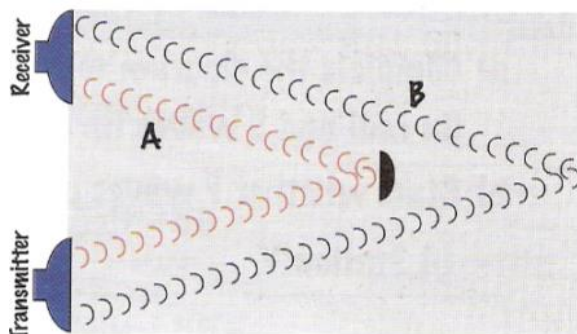
.....

.....

.....[3]



4. Ultrasound can be used to detect flaws in a material. An arrangement for this is shown below:



a) Which wave would arrive first to the receiver?

.....[1]

b) Explain how this arrangement works to detect flaws.

.....

.....

.....

.....[1]

5. Explain how ultrasound can be used by dentists to clean plaque from teeth.

.....

.....[1]

Sound Calculations

1. A spectator on a sports field is 200m from the start. She see the starting gun fire and then hears the sound of the shot 0.6s later.

a) Calculate the speed of the sound

.....[1]

b) What will be the wavelength if the frequency if the sound is 200Hz?

.....[1]

c) Another sound wave has a frequency of 2kHz. How does the frequency compare to the wave in part (b)? What w the ratios of their **wavelengths** be?

.....[1]

2. Sound travels through some things but not others

a) Which of the following can sound travel through? [4]

☐ Solids

☐ Vacuum

☐ Liquids

☐ Gases

b) Explain your answers

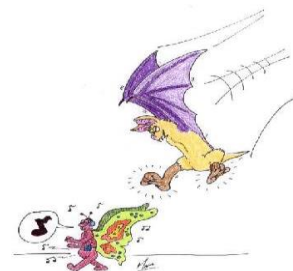
.....[1]

c) Which medium in a) does sound travel fastest in? Explain why

.....[2]

3. Bats use ultrasound to catch their prey. A typical victim would be a mosquito or a moth. A bat cannot sense anything smaller than the wavelength of sound it uses. What frequency does the bat need to send out to help catch a moth? (Hint: estimate the size of a moth first)

.....[3]



4. Some students are doing an experiment to measure the speed of sound. They are positioned at 200m intervals across the field. Each has a stopwatch. The student with the starting pistol simultaneously pulls the trigger and drops his arm. The students START their stopwatches when they see the arm fall, and STOP the watches when they hear the bang. Their times are shown below:

Distance (m)	200	400	600	800	1000
Time (s)	0.9	1.2	1.8	2.4	3.0

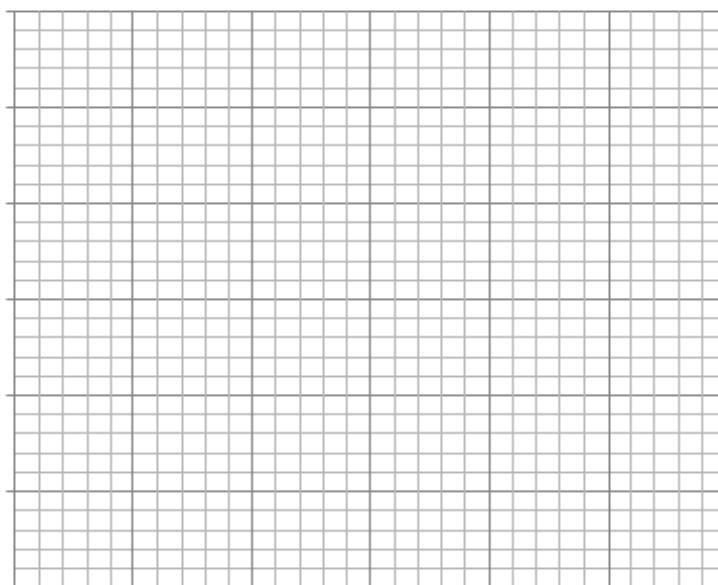
a) Plot a graph of distance against time [3]

b) Use your graph to calculate the speed of time

.....[2]

c) One point does not quite fit the pattern. Why might you expect this point to be the one least accurately measured?

.....[1]

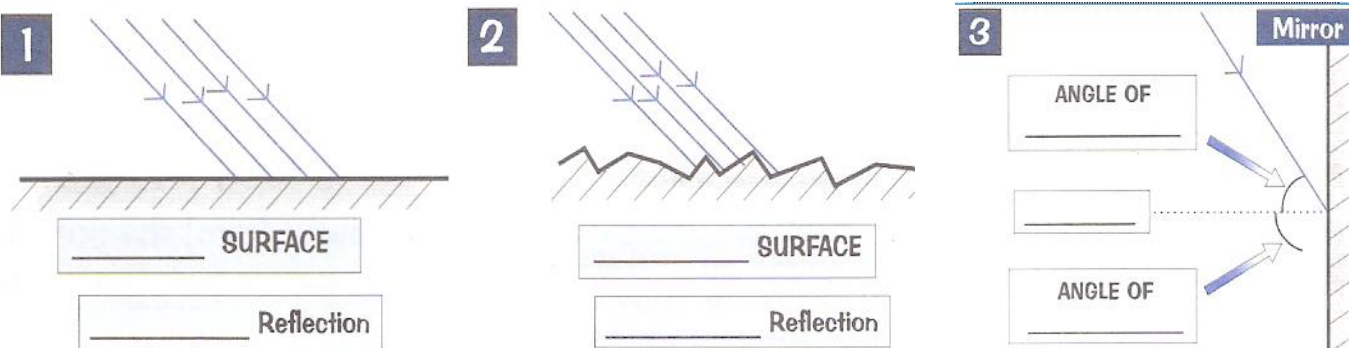


Reflection

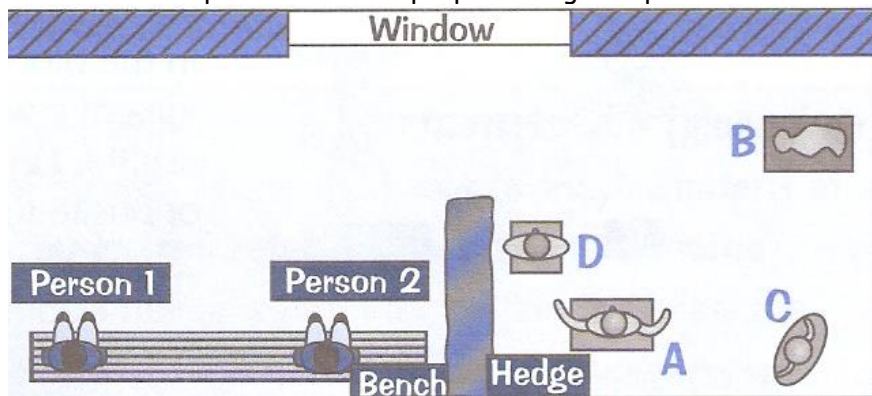
1. Fill in the blanks:

- Some objects give out their own light. These are called objects. We see all other objects because they light. These are called objects. [3]
- Some objects reflect light without sending it off in many different directions. This is called a reflection and objects which do this produce a [3]
- Most objects send the reflected light in many different directions giving a reflection. These objects do not produce a image [2]
- The law of reflection states that "the angle of is to the angle of [3]

2. The diagrams show rays arriving at a surface. Complete the labels and draw the reflected rays :



3. Look at the plan view of two people sitting on a park bench



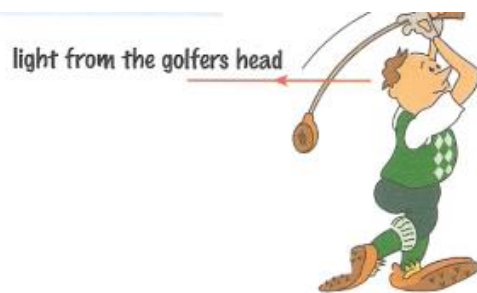
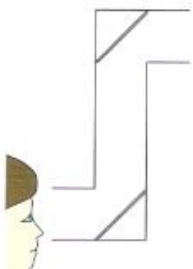
They can see some statues reflected in the window.

Use the law of reflection to decide which of the statues each can see

Person 1:.....[2]

Person 2:.....[2]

4. The diagram shows a periscope with two mirrors. Complete the path of the ray through the periscope.



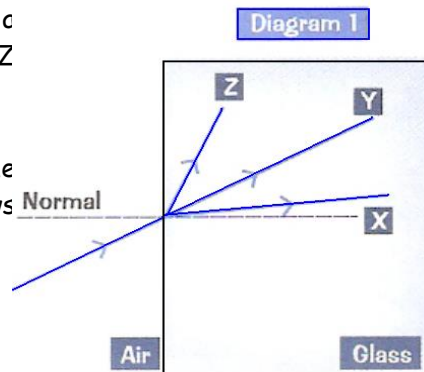
[1]

Refraction

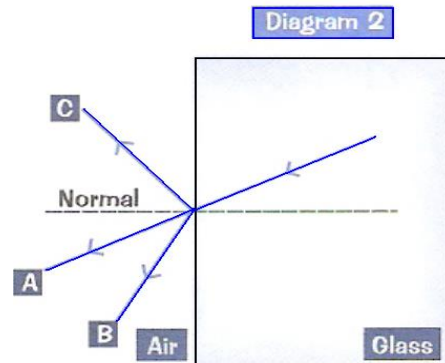
1. Fill in the blanks:

- Light travels at different in different media (e.g. air to glass)
- The speed of light will when it travels to a denser medium
- When the light goes back into air its speed will
- The normal is an imaginary line which is at to the surface
- The frequency of light when light travels through a denser medium [5]

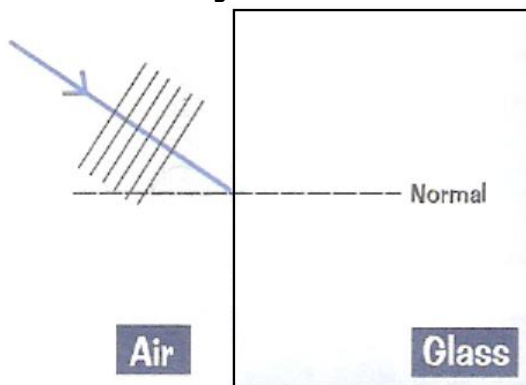
In the diagram 1, a ray **enters** a glass block. Which ray; X, Y or Z shows how it will continue? [1]



In the diagram 2 a ray **leaves** the block. Which ray; A, B or C shows the correct path?



- When a ray of light enters a glass block it (**slows down / speeds up**) and is bent (**towards / away from**) the normal.
- When a ray of light leaves a glass block it (**slows down / speeds up**) and is bent (**towards / away from**) the normal.
- Look at the diagram

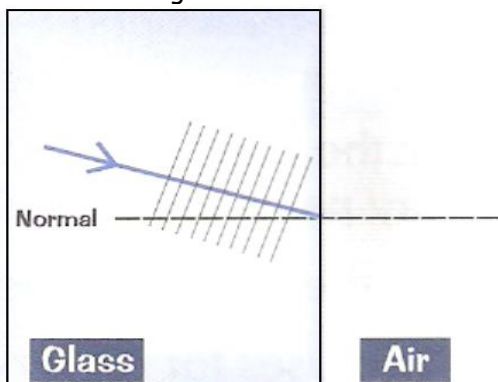


Write an explanation of the refraction of this ray in the block using the words:
wavefront, angle to the normal, slows down, wavelength, direction, frequency, unchanged

.....

[3]

- Look at the diagram



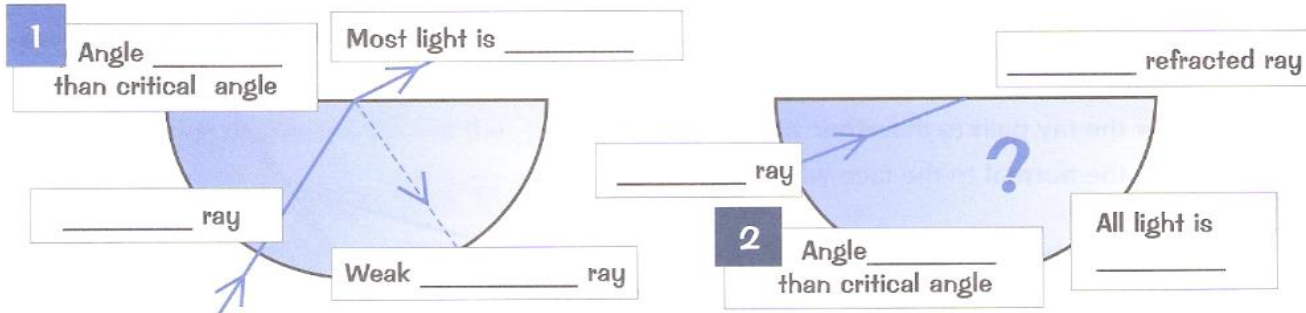
Write an explanation of the refraction of this ray as it exits the block using the words:
wavefront, angle to the normal, speeds up, wavelength, direction, frequency, unchanged

.....

[3]

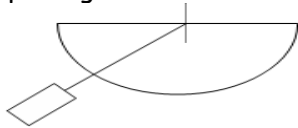
Total Internal Reflection

1. The diagram shows two identical glass blocks with a ray entering at two different angles



- Complete the labelling for diagram 1 and diagram 2. [8]
- Complete diagram 2 [2]

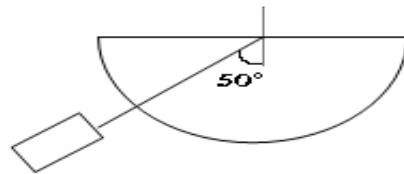
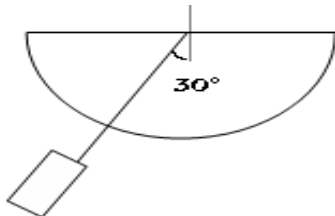
2. Joseph sets up the glass block shown below.



Look at the diagram and label :
the normal (N)
the angle of incidence (i)

- Explain why the ray of light passes straight through and is not deviated when entering the glass block[1]

- The critical angle of glass is 42° . The diagrams below show light incident at an angle of 30° in the first figure and 50° in the second figure. Complete the diagrams, showing the path the ray follows after it hits the straight edge. [2]



- Complete:
 - In the first figure the ray of light is[1]
 - In the second figure the ray of light is[1]

3. The following table of results is obtained from an experiment to find the critical angle of a semicircular plastic block.

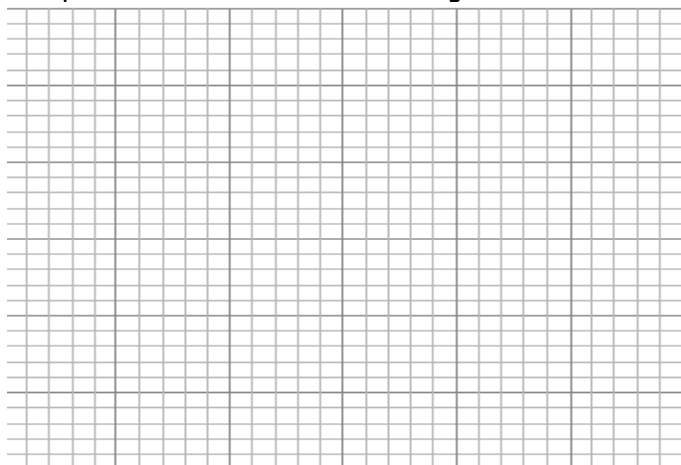
Angle in plastic							
Angle in air							

Plot a graph of angle in air (y-axis) against angle in plastic (x-axis). Draw the best smooth curve. [3]

From your graph find the angle in air when the angle in plastic is 35° [1]

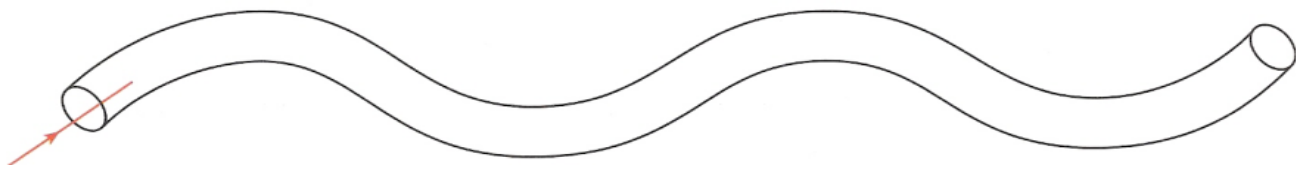
From your graph find the angle in plastic when the angle in air is 25° [1]

What is the critical angle of this kind of plastic? [1]



Uses of Total Internal Reflection

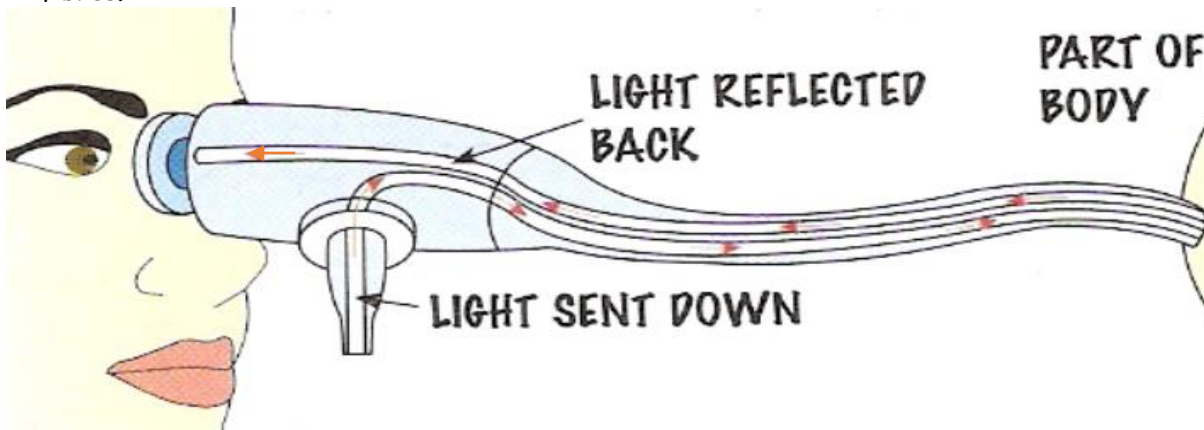
1. The diagram shows a length of optical fibre. Its critical angle is 42° .



- a) Complete the passage of the ray through the optical fibre. Make sure to mark the angles of incidence and reflection [2]
b) Fibre optics can be used for telecommunications. Explain the advantages over electrical signals

.....[2]

2. An endoscope is an instrument that can be used by doctors to see inside a patients knee. This instrument uses optical fibres.



Explain how the endoscope works:

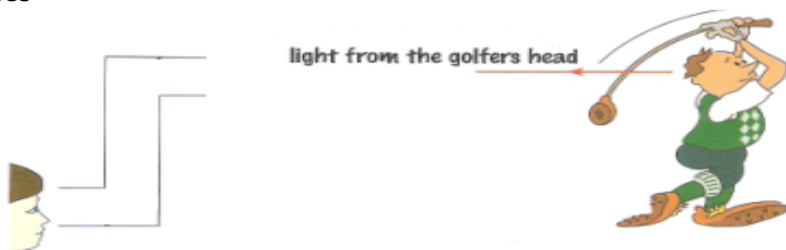
.....[2]

3. Reflecting prism are used in periscopes

Complete the diagram to show:

The arrangement of the prisms [1]

The path of the ray of light

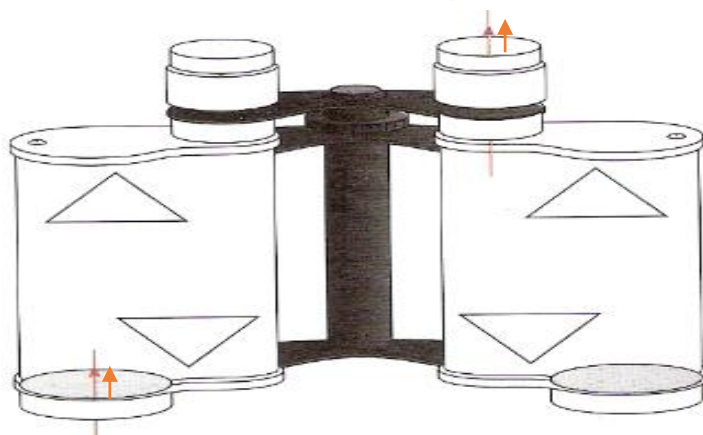


They are also used in prism binoculars.

Complete the path of light [2]

Explain why the light is reflected in the prism and not refracted, even though it is 'see through'

.....
.....
.....
.....[2]



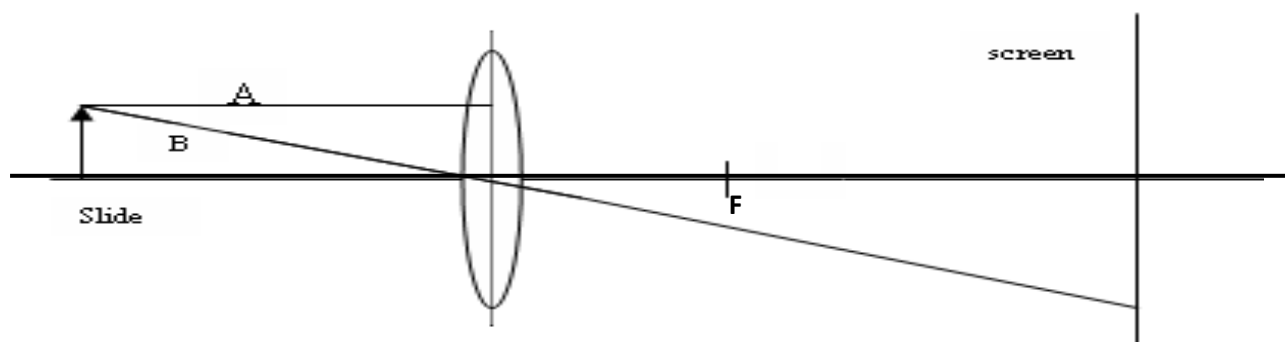
Lenses

1. An object of height 2 cm is placed 7 cm in front of a converging lens of focal length 3 cm. Draw a ray diagram to show this. Use an appropriate scale if needed. [3]

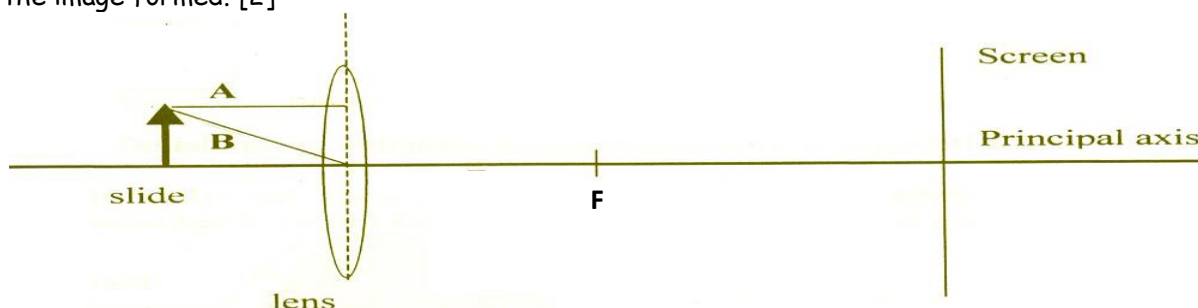
- a) What is the height of the image?[1]
- b) Name three properties of this image.[3]
- c) The object is moved 2cm from the lens. Draw the new ray diagram [3]

- d) Name three properties of this image.[3]
- e) Suggest one practical use of an image formed in this way by a converging lens.[1]

2. The figure below shows an incomplete ray diagram produced by a slide projector.



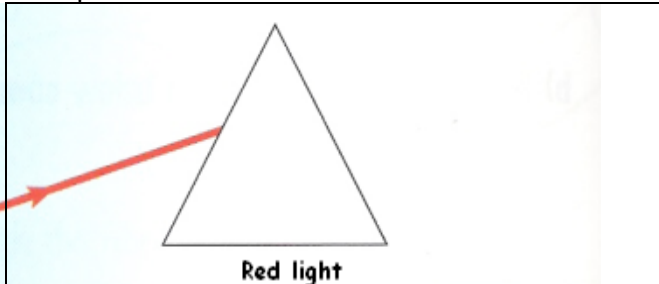
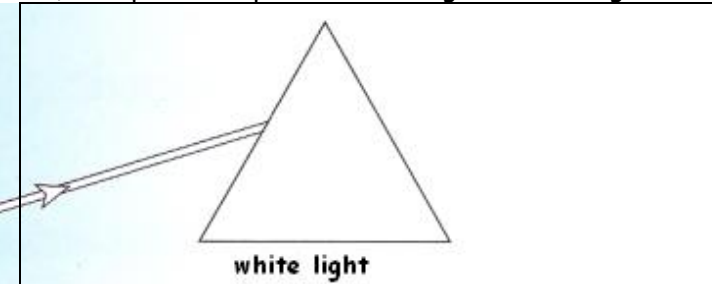
- Is the lens converging or diverging?[1]
- Draw the arrows on rays A and B, to show the direction followed by the light. [1]
- Complete ray A to show the image obtained on the screen. [1]
- Give three properties of the image.....[3]
- When the object was placed 8 cm in front of the lens, a focused image was obtained on a screen placed 2m behind the lens. Work out the magnification.
.....[2]
- A detail on the slide is 12 mm high. Work out the size of the image formed.
.....[2]
- The slide is now moved towards the lens as shown in the diagram below. Complete the rays A and B, and show the position of the image formed. [2]



Dispersion of Light

1. The drawing shows a triangular prism

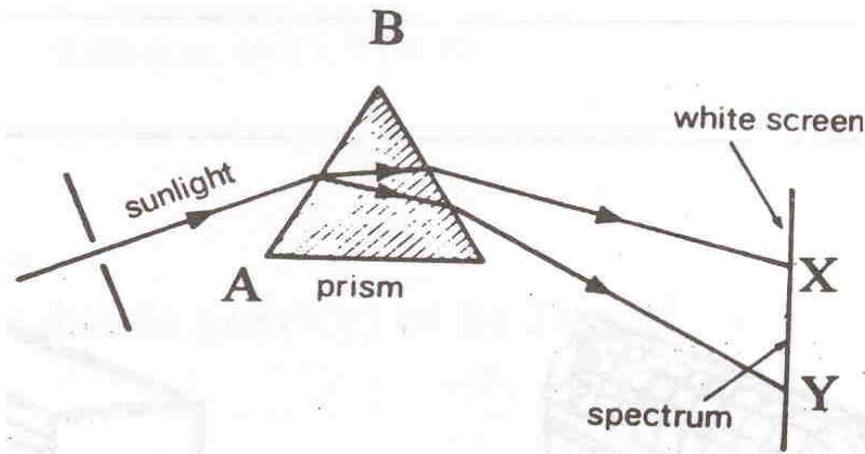
a) Complete the path of white light and red light through the prism



b) Explain why white light is split up by the prism

.....[2]

2. By placing a prism as shown in the diagram and shining a narrow beam of white light on the face AB, Sarah finds she can obtain faint colours on the screen.



a) Dispersion of white light results on the screen between X and Y. Explain the meaning of the word 'dispersion'.

.....[2]

b) Which colours are found at X and Y on the screen?

.....[2]

c) A type of invisible radiation can be found beyond the end Y of the spectrum. What is it called?

.....[2]